From Human Markets to Machine Markets: Transient Amplification, an Exact Depth Condition, and a Maximal Pre-Trade Guard

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Revised preprint, October 2026

Abstract

Eigenvalues and volatility do not bound the losses a stable market produces before it settles. We show that cumulative transient risk in a limit order book equals long-run impact divided by reversion net of momentum, that a pre-trade guard on this summed response is the largest safe admission rule without identifying traders, and that dealers under-supply reversion in equilibrium. The statistic forecasts next-hour adverse excursions beyond volatility forecasts in four pre-registered tests in thin Binance books and, more weakly, in thin US stocks; it is negligible on Bybit, and as a position limit it improves volatility limits only modestly.

Keywords: transient amplification, non-normal dynamics, limit order book, market resilience, liquidity externality, pre-trade risk control, pre-registration, replication.
JEL: G01, G10, G14, G18, C32, C58.

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Reproducibility. Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Appendix I maps each table, figure and number to its source. Pre-registered specifications, their SHA-256 hashes and test-set unlock counters are in the replication package. Every simulation number was re-run for this version.

1 Introduction

On October 10, 2025, crypto derivatives markets liquidated an estimated $19 billion of leveraged positions in a single day, the largest such event on record (21Shares, 2025); on May 6, 2010, US equity prices fell and recovered by several percent within minutes (Kirilenko et al., 2017). In both episodes the losses that mattered were transient: positions were closed out on the path, before prices settled. This paper is about that path risk in markets where algorithms trade with each other.

Algorithmic markets are closed-loop systems: trading moves prices, prices move algorithms, and algorithms trade. A common shortcut equates the stability of such a system with its eigenvalues lying inside the unit circle. That is the wrong object for a leveraged intermediary, whose survival depends on the path of losses before the system settles. A market can be spectrally stable and still push a dealer through its maintenance margin within seconds.

This paper asks two questions. When does a limit order book amplify a flow shock, so that prices keep moving after the shock has passed? And how can a dealer or a venue control transient loss without identifying who is trading, which the companion paper (Niroula and Niroula, 2026) shows is expensive or impossible from public data? We give exact answers in a linear book model and then confront them with data, reporting the results as they came out.

1. Spectral stability does not bound transient loss. The exact safe shock radius of a stable linear market is the loss budget divided by its peak transient gain, and that gain can be arbitrarily large at any fixed spectral radius (Theorem 3.1). Markets with identical eigenvalues, one-step volatility and stationary price variance can have arbitrarily large worst-case transient gain (Proposition 3.2). Supervision through eigenvalues or volatility alone is therefore insufficient in principle.

2. When a book amplifies. We distinguish peak overshoot, a response that rises above its direct impact, from cumulative amplification, a summed response far above it. When order flow depletes one side of the book and the book refills at rate r, the price response overshoots its direct impact if and only if $\varphi\varepsilon > (1-a)\kappa$ (Theorem 4.2). With momentum traders the one-period margin $C_3 = \varphi\varepsilon + \kappa^2\eta q - (1-a)\kappa$ is sufficient (Proposition 4.4)

but no one-period condition is exact (Remark 4.7); momentum alone cannot make a stable book overshoot (Theorem 4.5), and an explicit condition rules overshoot out (Theorem 4.6). Cumulative risk has a closed form, $S = \tilde{\kappa}/(m - q\tilde{\kappa})$: long-run impact divided by reversion net of momentum (Proposition 4.8). It needs no depletion, and it is the form our evidence supports: crypto books show no peak overshoot, but thin books revert slowly, so their summed response is large.

3. The maximal safe guard. For bounded shocks, worst-case loss over a horizon is inventory times the shock bound times the summed absolute impulse response, the classical ℓ_1 gain of control theory. A pre-trade guard on this summed gain is necessary and sufficient for solvency, so it admits the largest safe set of actions (Theorem 5.1). A guard on the peak response, the quantity closest to the eigenvalue intuition, admits unsafe actions. A Grönwall bound carries the guard to an estimated operator (Theorem 5.3), a contraction-block certificate makes it computable on a finite horizon (Lemma 5.2), and its inputs are identified from prices without identifying the algorithms (Propositions 2.2 and 5.4). We embed the guard in a dealer's balance sheet with margin, hedging and fill accounting (Theorem 5.5) and show that strategic re-optimization by filtering algorithms reinforces it (Proposition 5.7).

4. Who controls the risk. Reversion is a public good. In equilibrium dealers underprovide it, momentum traders over-provide feedback, fewer dealers mean more transient risk, and the rebate that restores the first best is $(1 - \theta/N)\Lambda S^2/\tilde{\kappa}$, a function of the guard's own statistic (Proposition 6.1).

5. Evidence. On Binance BTCUSDT futures the amplification statistic is negative on 93.5% of days, depth does not carry an overshoot, and the statistic fails a pre-registered forecasting test (Section 7); unit-free spot amplification is indistinguishable from one (Section 8). Across 30 perpetuals, Binance's depth bands, which are re-centred on the current mid, show taker flow replenishing the side it hits; on a fixed price grid in Bybit's full order books, the same flow depletes it, so the replenishment finding is an artefact of the band construction (Section 9.4). The amplifying regime is more frequent in thin books (Section 9). There, six-hour estimates of the amplification margin forecast the size of the next hour's adverse excursion out of sample in four pre-registered tests: on the original ten symbols (+1.2

points of R^2), on ten new symbols (+0.85), on fresh days with price-impact controls (+1.6), and on further fresh days beyond a full volatility-forecasting baseline (+2.8, with better tail classification). The summed response S, the guard's own statistic, carries the same information (+2.1 points beyond the volatility baseline), unit-free versions work, and the content comes mostly from impact-weighted reversion, as Proposition 4.8 implies. Within a volatility quintile, hours in the top quintile of the statistic have next-hour excursions about 26 basis points larger than hours in the bottom quintile, enough to roughly double the hourly liquidation probability of a 20× position. The effect is much weaker in mid-depth books and absent in BTCUSDT. On Bybit it fails with trade data alone; with order-book depth it meets a pre-registered criterion, but the gain is about one eightieth of Binance's (Table 26). In a regulated market, 23 thin US stocks with consolidated trades, the depth-free statistic passes a fifth pre-registered forecasting test (+0.50 points of R^2 beyond the same volatility baseline), again through impact-weighted reversion; unlike on Binance, mid-size stocks show a gain too (Table 18). As a position limit, a six-hour estimate of the response adds nothing to volatility, but once the operator is certified from the previous three days, scaling a volatility limit by the gain multiple lowers tail loss per unit of position by 2.4% in a fifth pre-registered test, a narrow pass; limits built on the estimated response alone remain worse than volatility (Section 9.14).

What we do not claim. We do not claim that crypto books amplify shocks through depth: flow depletes the side it hits on a fixed price grid, but we observe no peak overshoot, and the depth term carries no forecasting content. The thin-book forecasting result is reduced-form, and its size depends on the venue: largest in thin Binance books, about a fifth as large in thin US stocks, and negligible on Bybit; it improves tail classification only against the full volatility baseline. We do not claim that the guard replaces volatility in practical position sizing: its theorems are statements about worst-case loss under a certified operator, an estimated six-hour response adds nothing to volatility, and a three-day certified estimate improves a volatility limit only modestly (2.4%, a narrow pass) while the guard alone does worse. The October 10, 2025 episode is reported descriptively and supports no inference (Section 9.15). One test cell of the first thin-book test was executed twice with unchanged code and specification (Section 9.5). The equilibrium is a stylized model; thinner books do carry more cumulative risk per dollar of flow, as it predicts, but through larger impact rather

than through the weaker reversion it names.

1.1 Related literature

Closest work. Three strands are closest. Models of transient impact and resilience (Obizhaeva and Wang, 2013; Bouchaud et al., 2004, 2018) take the decay of impact as given; we derive when depletion, refill and momentum make a book overshoot, give cumulative impact in closed form, and ask who supplies the reversion that bounds it. Models of liquidity spirals (Brunnermeier and Pedersen, 2009; Shleifer and Vishny, 1992) explain how funding constraints amplify shocks; our amplification is a property of book dynamics and does not need funding constraints, and our guard is the pre-trade control that bounds the resulting loss. Sornette et al. (2023) show that non-normal interactions create transient bubbles; we give a necessary and sufficient pre-trade control for such transients and test the conditions for them in order-book data.

Market microstructure and resilience. Transient impact and resilience follow Obizhaeva and Wang (2013), Bouchaud et al. (2004) and Bouchaud et al. (2018); optimal execution with impact is in Almgren and Chriss (2001) and Cartea et al. (2015). Depth imbalance as a short-horizon price predictor is documented by Cont et al. (2014). Biais et al. (2005) survey the field; price impact as a measure of information and illiquidity goes back to Kyle (1985) and Amihud (2002). Inventory-based dealer models go back to Ho and Stoll (1981), and Avellaneda and Stoikov (2008) give the standard high-frequency market-making model. Empirically, resiliency, the speed at which a book refills after a liquidity shock, is measured by Biais et al. (1995), Degryse et al. (2005) and Large (2007), who estimates it with a self-exciting point process for order events; Foucault et al. (2005) show that it is set by the mix of patient and impatient traders and the tick size. Our reversion term m is the reduced-form counterpart of resiliency, and $S = \tilde{\kappa}/(m - q\tilde{\kappa})$ shows how it combines with impact and momentum: a book that is resilient in this sense can still carry a large cumulative response when momentum feedback absorbs most of its reversion. Our book model is a minimal two-state version of resilience dynamics, chosen so that the amplification condition is exact.

Feedback, fire sales and automated markets. Liquidity spirals and fire sales are in Shleifer and Vishny (1992) and Brunnermeier and Pedersen (2009); automated-market crashes in Kirilenko et al. (2017); and the market-design case for discrete auctions in Budish et al. (2015). Menkveld (2013) documents high-frequency market makers as the new intermediaries, and algorithmic trading has been found to improve liquidity and price discovery in normal times (Hendershott et al., 2011; Brogaard et al., 2014); our question is what happens on the path after a shock. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles in unleveraged systems. We add an exact condition for when a book amplifies, a control that does not need the identity of the traders, and evidence on whether the condition holds in a large market.

Early warning and liquidity crashes. The best-known early-warning metric for flow-driven crashes is VPIN (Easley et al., 2012), whose forecasting content beyond volume and volatility is disputed (Andersen and Bondarenko, 2014). Equilibrium models explain crashes through endogenous liquidity withdrawal (Huang and Wang, 2009) and illiquidity contagion across assets (Cespa and Foucault, 2014). Our statistic differs in kind: it is a structural quantity, the summed response of the book to a unit of flow, with a closed form and a control-theoretic meaning, and we evaluate it out of sample against a full volatility-forecasting baseline. In a post-hoc head-to-head check on the 20-symbol thin-book panel (`R8_vpin.json`), 50-bucket VPIN over the preceding six hours is negatively correlated with $\hat{C}_3$ (Spearman $\rho = -0.352$) and adds 1.24 percentage points of out-of-sample R^2 beyond the baseline, while $\hat{C}_3$ adds 2.03 percentage points (95% interval $[1.01, 3.42]$, Clark–West $t = 12.14$) on top of the baseline plus VPIN, retaining 73% of its unconditional R^2 gain (2.79 points). Crypto markets are studied as a laboratory for microstructure and arbitrage by Makarov and Schoar (2020).

Non-normal dynamics and robust control. Transient growth of stable non-normal systems and the Kreiss matrix theorem are classical (Trefethen and Embree, 2005; Kreiss, 1962); the sharp *en* constant is due to Spijker (1991). Standard H_∞ control bounds frequency-domain gain under known coordinates. The time-domain worst case under bounded disturbances, the ℓ_1 norm of the impulse response, is the object of ℓ_1-optimal control (Vidyasagar, 1986; Dahleh and Pearson, 1987; Dahleh and Diaz-Bobillo, 1995), and our guard's worst-

case identity is that classical fact. What we add is its use as an admission rule on a dealer's balance sheet, its invariance to the unobservable cohort coordinates and identification from prices, and its link to who supplies reversion in equilibrium. Cross-impact under no dynamic arbitrage is studied by Schneider and Lillo (2019).

Regulation. SEC Rule 15c3-5 requires broker-dealers with market access to enforce pre-trade controls on credit and capital (SEC, 2010). Our guard is a candidate pre-trade control with a provable maximality property.

Forecast evaluation. Our predictive tests use Clark and West (2007) and Diebold and Mariano (1995) with Newey and West (1987) standard errors and block resampling in the spirit of Künsch (1989), the volatility baseline follows Corsi (2009) and J.P. Morgan/Reuters (1996), local projections follow Jordà (2005), and we correct for multiple testing as in Holm (1979). Following the case for pre-registration in science (Nosek et al., 2018) and the concerns about multiple testing in financial economics (Harvey et al., 2016; Harvey, 2017), we pre-register specifications, count every evaluation of a held-out period, and report the failures with the successes.

1.2 Outline

Section 2 sets out the closed-loop market. Section 3 shows why spectral stability is not solvency. Section 4 derives the amplification condition. Section 5 constructs and analyzes the guard, and Section 6 asks who supplies reversion in equilibrium. Sections 7 and 8 report the BTCUSDT evidence, and Section 9 the thin-book program, its replications and its failures. Section 10 discusses policy, and Section 11 concludes.

2 The closed-loop market

2.1 Price formation

Time is discrete with bars of length Δ. Let p_t be the price, $p_t^* = p_{t-1}^* + \xi_t$ the unobserved fundamental and $x_t = p_t - p_t^*$ the pricing error. Competitive dealers post aggregate depth

$D > 0$, so price impact is $\kappa = 1/D$, and fundamental arbitrage removes a fraction $\kappa\varphi_f$ of the error each period. With algorithmic demand u_t and liquidity noise, the pricing error follows

$$x_{t+1} = ax_t + \kappa u_t + w_{t+1}, \qquad a = 1 - \kappa\varphi_f \in [0, 1). \tag{1}$$

Algorithmic cohorts trade $u_t = \sum_i q_i y_{i,t}$ with exponential memory $y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i x_{t+1}$, a reduced form that arises from steady-state Kalman filtering and regularized online gradient descent (Niroula and Niroula, 2026). Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_{t+1} = Jz_t + bw_{t+1}$ with

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1\kappa q_1 & \cdots & \eta_1\kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K\kappa q_1 & \cdots & 1 - \eta_K + \eta_K\kappa q_K \end{pmatrix}. \tag{2}$$

2.2 Stability and the role of depth

Proposition 2.1 (Stability with one cohort)**.** *With one cohort (q, η) and $a = 1 - \kappa\varphi_f$, the market is Schur stable if and only if*

$$q < \varphi_f, \qquad \kappa < \frac{4 - 2\eta}{(2-\eta)\varphi_f - \eta q} \ \text{ when } (2-\eta)\varphi_f > \eta q, \qquad |1 - \kappa\varphi_f|(1-\eta) < 1. \tag{3}$$

Higher impact κ (thinner depth) lowers price persistence a; feedback is destabilizing only when cohort aggressiveness q reaches the arbitrage strength φ_f.

The condition $q < \varphi_f$ is the familiar statement that momentum demand must be weaker than value arbitrage. The upper bound on κ rules out oscillatory instability when the book is very thin.

2.3 What an observer can use

The companion paper shows that prices identify only the merged rate–weight measure $M = \sum_i q_i\eta_i\delta_{1-\eta_i}$ of the cohorts. A pre-trade control must therefore depend on the market only through objects that are the same for every cohort representation of M.

Proposition 2.2 (Input–output invariance)**.** *Two populations with the same merged measure generate the same transfer function* $H(z) = \kappa/[z - a - \kappa G(z)]$ *and the same impulse coefficients* $g_k = c_{\text{price}}^{\top} J^k b$ *for every* $k \geq 0$*. Under any invertible change of state coordinates* $z \mapsto Tz$*, the Euclidean gain* $\|CT^{-1}(TJT^{-1})^k\|_2$ *depends on* T*, whereas the price path*

$$\Delta p_{t+k} = \sum_{j \geq 1} g_{k-1+j} u_{t-j} + \sum_{m=0}^{k-1} g_{k-1-m} u_{t+m}$$

and its worst case over $\|u\|_\infty \leq \bar{u}$ *depend only on* $\{g_k\}$ *and are invariant to both* T *and cohort splitting.*

Numerically, splitting a cohort of 1,000 random economies into two with the same memory rate leaves every impulse coefficient unchanged up to rounding ($\max_k |g_k(\mathcal{E}^A) - g_k(\mathcal{E}^B)| \leq 2.5 \times 10^{-12}$ over 200 lags; `recompute_oct20.py`). Proposition 2.2 is the reason we build the guard on impulse coefficients and avoid state norms in latent coordinates. When a state norm is needed, we fix units by observable calm-market innovation standard deviations, which is what makes the spot statistic of Section 8 unit-free.

Proposition 2.3 (Quote cost of ambiguity)**.** *For a one-unit ticket with adverse mark* $Z_\theta \sim N(\mu_\theta, \sigma_\theta^2)$ *and a CARA dealer with risk aversion* γ*, let* $c_\theta = \mu_\theta + \gamma\sigma_\theta^2/2$*. Under competition with zero robust certainty equivalent, the informed quote is* c_{θ_0} *and the ambiguous quote* $\sup_\theta c_\theta$*; with inverse demand* $(V_0 - V)/b$ *and marginal cost* c*, raising an interior quote from* s_0 *by* Δ *costs* $b(s_0 - c)\Delta + \frac{b}{2}\Delta^2$ *in surplus.*

Ambiguity about the algorithms is costly only when it changes the worst relevant cost. A guard that bounds worst-case transient exposure can therefore narrow the ambiguity premium; whether it does in a real market is an empirical question we do not answer here.

2.4 Discussion of the assumptions

Linearity and small shocks. All dynamics are linear. This is the right approximation for the transient response to flow shocks of ordinary size, which is what a pre-trade control must bound one step ahead; it is not a model of regime changes such as exchange outages, where nonlinear liquidity withdrawal dominates.

One book, one asset. Sections 4–5 treat a single instrument; Proposition 3.4 shows how directed cross-impact enlarges transient gains in a network. Cross-margined portfolios need the multi-asset operator, which we leave to future work.

Bounded shocks. The guard bounds losses for shocks in a declared envelope. Choosing the envelope is a risk-management decision; a larger envelope trades admitted flow for safety. The theorems hold for any envelope, and the empirical work does not depend on it.

Observables. Every input to the guard (impulse coefficients, an operator estimate and its confidence radius, inventory and margin) is observable to a dealer or venue. Nothing requires the identities or memory parameters of the algorithms.

3 Spectral stability is not solvency

Theorem 3.1 (Safe radius, Kreiss bounds and the failure of the spectral radius)**.** *Let* $z_t = J^t z_0$ *with* $\rho(J) < 1$*, loss budget* $B > 0$ *and loss output* C*.*

(i) The largest initial-state radius with $\sup_t \|CJ^t z_0\|_2 \le B$ *for every* $\|z_0\|_2 \le r$ *is*

$$r^*_{\text{exact}} = \frac{B}{M_C}, \qquad M_C \equiv \sup_{t\ge 0} \|CJ^t\|_2. \tag{4}$$

(ii) With $C = I$ *and the Kreiss constant* $\mathcal{K}(J) = \sup_{|z|>1}(|z|-1)\|(zI-J)^{-1}\|_2$*, the Kreiss matrix theorem gives* $\mathcal{K}(J) \le \sup_t \|J^t\|_2 \le e\,n\,\mathcal{K}(J)$ *for* $J \in \mathbb{R}^{n\times n}$*, hence*

$$\frac{B}{e\,n\,\mathcal{K}(J)} \;\le\; r^*_{\text{exact}} \;\le\; \frac{B}{\mathcal{K}(J)}. \tag{5}$$

(iii) For any $\rho \in (0,1)$ *and any* $B > 0$ *there is a family* J_θ *with* $\rho(J_\theta) = \rho$ *for all* θ *and an output* $C = (1,0)$ *such that* $r^*_{\text{exact}}(J_\theta) \to 0$ *as* $\theta \to \infty$*.*

Table 2 illustrates the theorem for $J_\theta = \left(\begin{smallmatrix} 0.8 & \theta \\ 0 & 0.4 \end{smallmatrix}\right)$: the spectral radius is 0.8 throughout, while the price-row gain rises from 1 to 120 and the safe radius falls by the same factor. The Kreiss constant tracks the gain within the factor en.

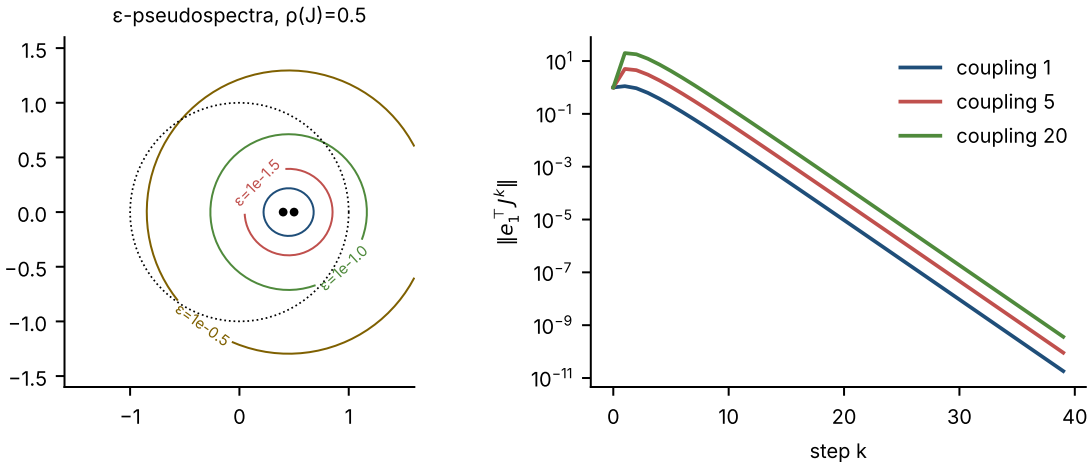


Figure 1: Left: ε-pseudospectra of $J = \left(\begin{smallmatrix} 0.5 & 5 \\ 0 & 0.4 \end{smallmatrix}\right)$; the eigenvalues (dots) are well inside the unit circle (dotted), but the $\varepsilon = 10^{-1}$ contour reaches beyond it. Right: price-row gain $\|e_1^\top J^k\|$ for couplings 1, 5 and 20.

The theorem concerns a specified initial-state ball and output norm. Changing state coordinates changes the disturbance set, so latent-state units cannot be read as monetary losses; this is why Section 5 works with input–output gains instead.

Proposition 3.2 (Spectral blindness; companion paper)**.** *For any eigenvalues* $0 < \lambda_2 < \lambda_1 < 1$*, one-step price variance and sufficiently large stationary price variance, there are two-state markets sharing all three quantities whose worst-case transient gain is arbitrarily large.*

At $\lambda = (0.8, 0.7)$ with one-step variance 1 and stationary variance 5, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 as the off-diagonal coupling rises from 1 to 100. Figure 1 shows the geometry for a constructed matrix with spectral radius 0.5: its pseudospectra cross the unit circle, and the price-row gain grows with the coupling before decaying.

Lemma 3.3 (No amplification without depth; companion paper)**.** *In the cohort economy* (2) *with nonnegative weights and* $\rho(J) < 1$*, the price impulse response is a nonnegative mixture of decaying exponentials with total weight* κ*. It is completely monotone and peaks at the direct impact.*

Lemma 3.3 is a negative result with a positive use. The cohort matrix (2) is non-normal, yet its price output never amplifies. Non-normality is necessary for price amplification but not sufficient, and adaptive momentum feedback alone does not supply it. Section 4 shows that depth depletion does.

3.1 Multi-asset coupling

Market makers trade many correlated instruments. Let $x_t \in \mathbb{R}^d$ be pricing errors and $u_t \in \mathbb{R}^d$ flows, with $x_{t+1} = Ax_t + Ku_t + w_{t+1}$, $A = \mathrm{diag}(a_1, \ldots, a_d)$, a cross-impact matrix K, and momentum flow $y_{t+1} = (I - H)y_t + Hx_{t+1}$, $u_t = Qy_t$. The stacked Jacobian is

$$J_{\text{multi}} = \begin{pmatrix} A & KQ \\ HA & I - H + HKQ \end{pmatrix} \in \mathbb{R}^{2d \times 2d}. \tag{6}$$

Proposition 3.4 (Directed cross-impact shrinks the safe radius)**.** *Suppose a leader–satellite structure ($K_{12} = 0$, $K_{21} \geq 0$) with $A, Q, K \geq 0$ entrywise, $H \in (0, 1)^d$, stable diagonal blocks and a fixed output $C \geq 0$. Let J_{diag} be the decoupled Jacobian with $K_{21} = 0$. Then every power J^t_{multi} is entrywise non-decreasing in K_{21}, so $M_C(J_{\text{multi}}) \geq M_C(J_{\text{diag}})$ and $r^*_{\text{exact}}(J_{\text{multi}}) \leq r^*_{\text{exact}}(J_{\text{diag}})$; likewise $\mathcal{K}(J_{\text{multi}}) \geq \mathcal{K}(J_{\text{diag}})$.*

The proposition concerns nonnegative downstream coupling under a fixed output and shock set. Hedged or signed monetary outputs need separate treatment, and the size of the effect requires an empirical calibration we do not provide. Persistence and transient amplification are nevertheless different things: in a network of identical cohort economies coupled by symmetric nonnegative cross-impact, which stays symmetric in suitable coordinates, cross-impact of 0.05 raises the spectral radius from 0.822 ($d = 1$) to 0.973 ($d = 10$) while the Kreiss constant stays at one (`recompute_oct20.py`). Directed coupling, as in Proposition 3.4, is what turns persistence into amplification.

4 When does a book amplify? An exact condition

Transient path risk takes two forms, and the distinction organizes both the theory and the evidence. The first is *peak overshoot*, a response that rises above its direct impact; Theorem 4.2 shows that it requires depth depletion to outweigh impact-weighted reversion. The

second is *cumulative amplification*, a summed response S much larger than direct impact; Proposition 4.8 shows that it arises without any depletion whenever reversion is slow relative to impact and momentum. The guard of Section 5 bounds the second. The data of Sections 7–9 show no peak overshoot in crypto books, even though flow depletes the side it hits on a fixed price grid, and they support the second form in thin books.

4.1 A book with depletion and refill

Let x_t be the price displacement, d_t the signed depth imbalance (bid minus ask depth near the mid, relative to total) and u_t signed order flow. Flow moves the price directly and depletes one side of the book, which refills at rate r; depth imbalance moves the price:

$$x_t = a\, x_{t-1} + \varphi\, d_{t-1} + \kappa\, u_t, \tag{7}$$

$$d_t = (1 - r)\, d_{t-1} + \varepsilon\, u_t, \tag{8}$$

with $\kappa > 0$, $a \in [0, 1)$ and $r \in (0, 1]$. Write $b = 1 - r$. With the sign convention $d = (B - A)/(B + A)$, a buy that consumes ask depth raises d, so $\varepsilon > 0$ means *depletion* of the side that is hit, and $\varphi > 0$ means that a bid-heavy book pushes the price up. The impulse response to a unit flow shock at $t = 0$ is $h_0 = \kappa$ and

$$h_k = a\, h_{k-1} + \varphi\varepsilon\, b^{k-1}, \qquad k \geq 1. \tag{9}$$

Definition 4.1. The book *amplifies* if $\max_k h_k > h_0$: the price keeps moving in the direction of the shock after the shock has passed.

4.2 The two-state theorem

Theorem 4.2 (Amplification condition)**.** *In model (7)–(8), the book amplifies if and only if*

$$C_2 \equiv \varphi\varepsilon - (1 - a)\kappa > 0.$$

When $C_2 \leq 0$ the response peaks at impact, $\max_k h_k = \kappa$.

Proof. If. $h_1 - h_0 = a\kappa + \varphi\varepsilon - \kappa = C_2 > 0$.

Only if. Suppose $C_2 \le 0$. If $\varphi\varepsilon < 0$, then by (9) $h_k = a^k\kappa + \varphi\varepsilon \sum_{j<k} a^{k-1-j} b^j \le a^k\kappa \le \kappa$. If $\varphi\varepsilon \ge 0$, let $\Delta_k = h_k - h_{k-1}$. Then $\Delta_1 = C_2 \le 0$ and, from (9),

$$\Delta_{k+1} = a\Delta_k + \varphi\varepsilon(b^k - b^{k-1}) = a\Delta_k - \varphi\varepsilon\, r\, b^{k-1} \le a\Delta_k.$$

By induction $\Delta_k \le 0$ for all k, so h is non-increasing and peaks at $h_0 = \kappa$. □

Interpretation. The left side $\varphi\varepsilon$ is the extra price pressure that a unit of flow creates one period later through the book it leaves behind. The right side $(1-a)\kappa$ is the part of the direct impact that reverts in one period. The book amplifies exactly when the depth channel adds more than mean reversion removes. Refill r does not enter the condition; it governs how long an overshoot lasts.

Proposition 4.3 (Shape of the overshoot)**.** *For $a \neq b$ the response is $h_k = Aa^k + Bb^k$ with $B = \varphi\varepsilon/(b-a)$ and $A = \kappa - B$. When the book amplifies ($\varphi\varepsilon > (1-a)\kappa$):*

(i) In the interior regime $0 < a \neq b < 1$ with $AB < 0$, the continuous profile $k \mapsto Aa^k + Bb^k$ on $k \ge 0$ has a unique stationary point at

$$k^* = \frac{\log\big(-A\log a/(B\log b)\big)}{\log(b/a)},$$

so the discrete peak lag is $\lfloor k^* \rfloor$ or $\lceil k^* \rceil$ when $k^* \ge 1$, and is 1 when $k^* < 1$ (because $h_1 > h_0$).

(ii) On the boundary cases: if $a = 0$ and $b \in (0,1)$, then $h_0 = \kappa$ and $h_k = \varphi\varepsilon b^{k-1}$ for $k \ge 1$, which peaks at $k = 1$; if $0 < a < b < 1$ and $A \le 0$, then $h'(k)$ turns negative at $k^* \le 1$ so the discrete peak is at $k = 1$; and in the confluent limit $a = b \in (0,1)$, $h_k = (\kappa + \varphi\varepsilon k/a)a^k$ with unique continuous peak at $k^* = 1/\log(1/a) - \kappa a/(\varphi\varepsilon)$.

In all cases with $a \in [0,1)$ and $r = 1 - b \in (0,1)$, the cumulative response is $\sum_{k\ge 0} h_k = \kappa/(1-a) + \varphi\varepsilon/[(1-a)r]$, and faster refill r lowers both the peak and the cumulative response.

We verify Theorem 4.2 in 1,995 random parameter draws and the refill comparative static in 565 amplifying draws, with no exceptions. The peak lag lies in the stated bracket in all of 150,039 amplifying draws; it is not always the integer nearest to k^*, which an earlier version

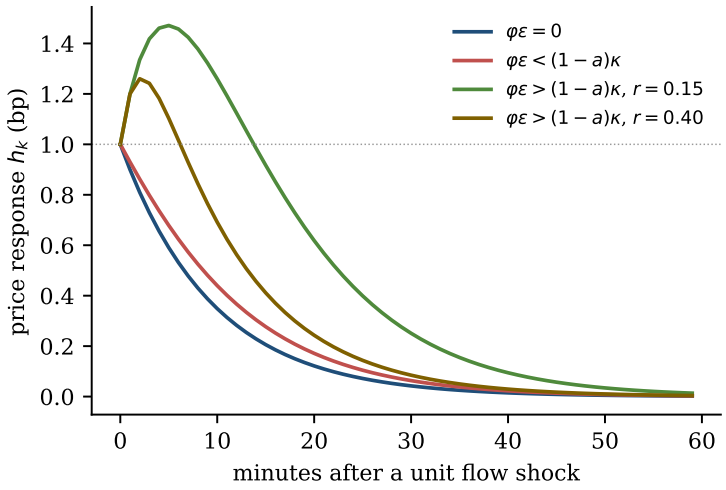


Figure 2: Illustrative impulse responses of model (7)–(8) ($a = 0.9$, $\kappa = 1$, $\varepsilon = 0.1$). The response overshoots only when $\varphi\varepsilon > (1-a)\kappa$; faster refill r lowers the overshoot. Parameters are illustrative, not estimates.

claimed and which fails in 15% of draws (`recompute_audit.py`). Figure 2 illustrates the four cases.

4.3 Momentum traders

Add momentum traders whose flow $f_t = (1-\eta)f_{t-1} + \eta q\, x_{t-1}$ enters both (7) and (8) alongside u_t.

Proposition 4.4 (One-period amplification margin)**.** *Let $C_3 \equiv \varphi\varepsilon + \kappa^2\eta q - (1-a)\kappa$. Then $h_1 - h_0 = C_3$, so $C_3 > 0$ implies that the book amplifies.*

Proof. With a unit shock at $t = 0$, $x_0 = \kappa$, $d_0 = \varepsilon$ and $f_1 = \eta q\kappa$, so $h_1 = a\kappa + \varphi\varepsilon + \kappa^2\eta q$. □

With two slow modes the converse fails, and no condition on the one-period margin is exact. Three results describe what can be said instead.

Theorem 4.5 (Momentum alone cannot make a stable book amplify)**.** *Suppose $\varphi\varepsilon = 0$ and $q \geq 0$, and the system is stable. Then $h_k \leq \kappa\,\rho^k < \kappa$ for every $k \geq 1$, where $\rho < 1$ is the spectral radius. In particular $C_3 < 0$.*

Proof. Without the depth channel the state (x_t, f_t) evolves by $M = \begin{pmatrix} a+\kappa\eta q & \kappa(1-\eta) \\ \eta q & 1-\eta \end{pmatrix}$, which is entrywise nonnegative, and $h_k = \kappa(M^k)_{11}$. The off-diagonal product is nonnegative, so the eigenvalues are real, and by the Perron–Frobenius theorem $\lambda_2 \le m_{11} \le \lambda_1 = \rho$. Writing $(M^k)_{11} = c_1\lambda_1^k + c_2\lambda_2^k$ with $c_1 = (m_{11}-\lambda_2)/(\lambda_1-\lambda_2) \in [0,1]$ and $c_2 = 1-c_1$ gives $(M^k)_{11} \le \rho^k$. Since $m_{11} \le \rho < 1$, $C_3 = \kappa(m_{11}-1) < 0$. □

Theorem 4.6 (A sufficient condition for no overshoot)**.** *For any signs of φ, ε and q, if*

$$\bar{C}_3 \equiv |\varphi\varepsilon|\Big(1 + \frac{\kappa|q|}{r}\Big) + \kappa^2|q| - (1-a)\kappa \;\le\; 0,$$

then $\sup_k |h_k| = \kappa$*: the response never exceeds its impact in either direction. When $\varphi\varepsilon \ge 0$ and $q \ge 0$, $C_3 > 0 \Rightarrow$ amplification $\Rightarrow \bar{C}_3 > 0$.*

Proof. Suppose $|x_j| \le \kappa$ for $j < k$. Then $|f_j| \le |q|\kappa$ for $j \le k$ and $|d_{k-1}| \le |\varepsilon|\big(b^{k-1} + \sum_{j\ge1} b^{k-1-j}|q|\kappa\big) \le |\varepsilon|(1 + |q|\kappa/r)$, so $|x_k| \le a\kappa + |\varphi\varepsilon|(1 + \kappa|q|/r) + \kappa^2|q| \le \kappa$. Induction from $x_0 = \kappa$ completes the proof. □

Remark 4.7 (Counterexamples to an exact one-period condition). Between C_3 and $\bar{C}_3$ the sign of C_3 does not decide. With $\kappa = 1$, $a = 0.98526$, $\varphi = 0.7409$, $\varepsilon = 0.01183$, $r = 0.03436$, $\eta = 0.5092$ and $q = 0.01127$ (all channels reinforcing, $\rho = 0.9995$), $C_3 = -0.00024$ but the response peaks at lag 60 at 1.092κ. With replenishment ($\varphi = 2.614$, $\varepsilon = -0.405$, $a = 0.9426$, $r = 0.033$, $\eta = 0.0016$, $q = 0.0785$, $\rho = 0.998$), $C_3 = -1.11$ yet the response oscillates to 9.2κ at lag 90. In a search over 400,000 parameter draws such cases occur only near the stability boundary or with slow refill and slow momentum.

Proposition 4.8 (Summed response in closed form)**.** *Let $\tilde{\kappa} = \kappa + \varphi\varepsilon/r$ (long-run impact: direct plus depth) and $m = 1 - a$ (reversion). In the stable three-state model,*

$$S \equiv \sum_{k\ge0} h_k = \frac{\tilde{\kappa}}{m - q\tilde{\kappa}},$$

and when $\varphi\varepsilon \ge 0$ and $q \ge 0$ the system is stable only if $m > q\tilde{\kappa}$. When all $h_k \ge 0$, S is the summed absolute response that enters the guard of Section 5.

Proof. $S = e_1^\top (I - M_3)^{-1}(\kappa, \varepsilon, 0)^\top$ with M_3 the transition matrix of (x, d, f). Direct computation gives $\det(I - M_3) = \eta\,[\,r(m - \kappa q) - \varphi\varepsilon q\,]$ and the stated ratio; the formula matches the simulated sums to a relative error of 7×10^{-14} in 3,000 stable draws. □

Cumulative transient risk is long-run impact divided by reversion net of momentum. Two practical consequences follow. First, the statistic that matters for losses is the summed response, which grows without bound as reversion falls to $q\tilde{\kappa}$, even when no single lag overshoots (Theorem 4.5). Second, C_3 is best read as a one-period amplification margin: it moves with m, q and $\tilde{\kappa}$ in the same direction as S and is what we estimate in Sections 7 and 9. Both statistics are invariant in sign to rescaling flow.

5 An identification-free pre-trade guard

5.1 The quantity gate

Observable dynamics are $z_{j+1} = Jz_j + Hv_j$ with loss output $l_t = Cz_t$ and impulse coefficients $g_m = c_{\text{price}}^{\top} J^m H$. Let $b_t = CJ^t z_0$ be the drift from past flow, $G_{\text{single},t} = \max_{m<t} |g_m|$ the single-shock peak gain up to t, and $G_{\text{cum},t} = \sum_{m=0}^{t-1} |g_m|$ the cumulative convolution gain for repeated per-period trades $|v_j| \le V$. Superposition gives, under a reserve L_{other} and execution cost c_{exec}, the sharp repeated-trade allowance (and its single-impulse counterpart)

$$\begin{aligned} V^*_{\text{repeated}} &= \min_{t:\, G_{\text{cum},t}+c_{\text{exec}}>0} \frac{B - L_{\text{other}} - |b_t|}{G_{\text{cum},t} + c_{\text{exec}}}, \qquad \bar{R} = V^*_{\text{repeated}}/\tau, \\ V^*_{\text{single}} &= \min_{t:\, G_{\text{single},t}+c_{\text{exec}}>0} \frac{B - L_{\text{other}} - |b_t|}{G_{\text{single},t} + c_{\text{exec}}} \end{aligned} \tag{10}$$

over a window of τ seconds. Using $G_{\text{single},t}$ in place of $G_{\text{cum},t}$ when trades repeat every period under-states worst-case cumulative exposure whenever multiple lags are nonzero. The gate needs only impulse coefficients and the current drift.

5.2 Maximality

A dealer with inventory I chooses an admission level $\alpha \in [0, 1]$ (the fraction of incoming algorithmic flow it accepts, or the depth it quotes). The closed-loop operator is $J(\alpha) = J_0 + \alpha B_F$ with price impulse response $g_k(\alpha)$. Shocks satisfy $|u_t| \le \bar{u}$. Over horizon H the price change is $\sum_{k<H} g_k(\alpha) u_{t-k}$, and the dealer has a loss budget $\mathcal{B}$.

Theorem 5.1 (Summed-gain guard is maximal)**.** *The worst-case loss over horizon H is*

$$L_{\max}(\alpha) = |I(\alpha)|\,\bar{u}\sum_{k<H}|g_k(\alpha)|.$$

Hence the guard "admit α iff $L_{\max}(\alpha) \le \mathcal{B}$" is necessary and sufficient for bounded-shock solvency: it admits every safe α and no unsafe one. Replacing $\sum_k |g_k|$ by $\max_k |g_k|$ can admit unsafe levels; replacing it by $H \max_k |g_k|$ is safe but conservative.

Proof. By Hölder's inequality $|\sum_k g_k u_{t-k}| \le \bar{u}\sum_k |g_k|$, with equality for $u_{t-k} = \bar{u}\,\mathrm{sign}(g_k)$, which is admissible. Multiplying by $|I(\alpha)|$ gives $L_{\max}$, so the admission set equals the safe set. Since $\max_k |g_k| \le \sum_k |g_k| \le H \max_k |g_k|$, the peak guard under-states loss and the H-scaled guard over-states it. □

The worst-case identity in Theorem 5.1 is not new as mathematics: the induced ℓ_∞ gain of a linear convolution operator is the ℓ_1 norm of its kernel, the fact on which ℓ_1-optimal control of persistent bounded disturbances is built (Vidyasagar, 1986; Dahleh and Pearson, 1987; Dahleh and Diaz-Bobillo, 1995). We use it rather than claim it. The finance content lies elsewhere: the dealer's admission level changes the operator itself, so the guard is a rule over an action set rather than a controller design; the bound is carried through margin, hedging and fill accounting (Theorem 5.5); its single input is invariant to how the unobserved traders are split into cohorts and is identified from prices (Propositions 2.2 and 5.4); the peak-response and eigenvalue rules it is compared with are the ones closest to current supervisory practice; and the same statistic prices the corrective rebate in equilibrium (Proposition 6.1).

Numerically, the guard coincides with a brute-force adversary's maximal safe α in 200 of 200 random states. The peak-gain guard admitted an unsafe level in 2,000 of 2,000 states, the H-scaled guard was conservative in all 2,000, and $L_{\max}$ was non-decreasing in α in all 2,000, so the admission set was an interval in these examples.

5.3 Computability: a finite-horizon certificate

Lemma 5.2 (Contraction-block tail bound)**.** *Let $\rho(J(\alpha)) < 1$. Suppose that at horizon H there is a block length $H_0 \le H$ with $q(\alpha) \equiv \|J(\alpha)^{H_0}\|_2 < 1$, and let $\bar{M}_{H_0}(\alpha) =$*

$\max_{0\le r<H_0} \|c_{\text{price}}^\top J(\alpha)^r\|_2$ *and* $M_{C,H}(\alpha) = \max_{k\le H} \|c_{\text{price}}^\top J(\alpha)^k\|_2$. *Then*

$$\sup_{k\ge 0} \|c_{\text{price}}^\top J(\alpha)^k\|_2 \le \max\Big(M_{C,H}(\alpha),\ \bar{M}_{H_0}(\alpha) q(\alpha)^{\lfloor (H+1)/H_0 \rfloor}\Big), \tag{11}$$

$$G_{\text{cum},\infty}(\alpha) \le G_{\text{cum},H}(\alpha) + \frac{\bar{M}_{H_0}(\alpha) H_0\, q(\alpha)^{\lfloor (H+1)/H_0 \rfloor}}{1 - q(\alpha)}.$$

Proof. Write any $k > H$ as $k = mH_0 + r$ with $m \ge \lfloor (H+1)/H_0 \rfloor$ and $0 \le r < H_0$. Submultiplicativity gives $\|c^\top J^k\| \le \|c^\top J^r\|\, \|J^{H_0}\|^m \le \bar{M}_{H_0} q^m$. Taking the supremum and summing the geometric tail gives (11). □

The certificate turns an infinite-horizon supremum into a finite computation with a verified remainder. If the second term in the maximum is below $M_{C,H}$, truncation at H is exact.

5.4 Robustness to estimation error

Theorem 5.3 (Robust gain)**.** *Let* $M = \sup_k \|J^k\|$ *and* $S = \sum_k \|J^k\|$. *If* $\|E\| \le \epsilon$ *and* $\epsilon S < 1$, *then* $\sup_k \|(J+E)^k\| \le M/(1-\epsilon S)$.

Proof. Let $P_k = (J+E)^k$. Expanding, $P_k = J^k + \sum_{j<k} J^{k-1-j} E P_j$. With $m_k = \|P_k\|$, $m_k \le M + \epsilon \sum_{j<k} \|J^{k-1-j}\| m_j \le M + \epsilon S \max_{j<k} m_j$. Since $m_0 = 1 \le M$, induction gives $m_k \le M/(1-\epsilon S)$ for all k. □

The bound held in all 30,000 random perturbations of 100 non-normal operators, with median tightness 0.50 (realized gain over bound; maximum 0.57). In a simulated window estimator of the kind used in Section 8 (600 transitions), a residual-bootstrap 95% ball for the operator covered the true J in 94.7% of 300 replications, with median radius 0.086; but $\epsilon S < 1$ held in only 8% of windows. The certificate is valid but often unavailable at realistic sample sizes; longer windows or pooled estimates are needed to use it in practice.

Proposition 5.4 (No identification needed; numerical)**.** *In a two-cohort economy with rates* $0.7 \pm \Delta$, *the relative error of the estimated summed gain is flat in* Δ *(log–log slope* -0.15*), while the error of the cohort weights grows as* $\Delta^{-3.07}$.

The guard needs the summed gain, which is cheap to estimate and invariant to cohort splitting (Proposition 2.2); it does not need the algorithms' identities, which are expensive (Niroula and Niroula, 2026).

5.5 A calibrated illustration

At the median BTCUSDT depth estimates, price impact is so persistent ($\hat{a} \approx 0.97$ per minute) that the summed gain over 15 minutes is about 13 times the peak, so a guard calibrated to the peak would admit inventory about 13 times larger than is safe against a sustained flow of the declared size, even though this book does not amplify. The Internet Appendix gives the calculation (Figure 6, Table 24).

5.6 Two deployment modes

Mode A. *An individual dealer.* A price-taking market maker with a small share of aggregate depth scales its own quote size by α. This scales its passive fill intake but leaves aggregate depth, impact and the market operator unchanged.

Mode B. *A venue or sponsoring broker.* A gate at a matching engine or a market-access gateway (the setting of Rule 15c3-5) admits a fraction α of incoming aggressive algorithmic flow while resting depth stays posted. The operator becomes $J(\alpha) = A_0 + \alpha B_F$.

Algorithm (pre-trade guard). At each step, using only pre-shock information:

1. For each α on a grid $\mathcal{A} = \{0, 0.025, \ldots, 1\}$, compute $\rho(J(\alpha))$, the peak and summed gains $M_{C,H}(\alpha)$ and $G_{\text{cum},H}(\alpha)$, and the tail certificate of Lemma 5.2; if an operator confidence radius ϵ is available, inflate by Theorem 5.3.
2. Compute post-hedge and worst-case post-fill inventory, hedging cost and the post-fill margin increment.
3. Admit the largest $\alpha \in \mathcal{A}$ that satisfies the solvency inequalities of Theorem 5.5; if none does, set $\alpha = 0$, flag that no safe action exists, and trigger inventory reduction.

For low-dimensional J the grid evaluation is a sequence of matrix–vector products; execution takes well under one millisecond in compiled code, which we regard as an engineering detail rather than a result.

5.7 Balance-sheet solvency

Theorem 5.5 (Conditional bounded-shock forward solvency)**.** *Consider a dealer with cash K_t, inventory I_t, mark p_t, equity $E_t = K_t + I_t p_t$, maintenance margin $MM_t = \max(0, \mu|I_t|p_t - c_0)$ and remaining buffer $B_{\mathrm{rem},t} = E_t - MM_t$, a safety reserve R_{safe}, a hedge $\Delta I_t^{\mathrm{hedge}}(\alpha)$ with cost $C_{\mathrm{hedge},t}(\alpha)$, worst-case passive fills $\Delta I_{\mathrm{max},t}^{\mathrm{fill}}(\alpha)$ with margin increment $C_{\mathrm{fill},t}(\alpha)$, and declared one-step and cumulative shock envelopes $\bar{u}_{\mathrm{step},t} \le \bar{u}_{\mathrm{cum},t}$. Let $|I_{t+1}^{\mathrm{post}}(\alpha)| = |I_t + \Delta I_t^{\mathrm{hedge}}(\alpha)| + \Delta I_{\mathrm{max},t}^{\mathrm{fill}}(\alpha)$. Suppose the guard admits α_t only if*

$$|I_{t+1}^{\mathrm{post}}(\alpha_t)|(1+\mu)\Big(M_C(\alpha_t)\|z_t\|_2 + s(\alpha_t)G_{\mathrm{cum}}(\alpha_t)\bar{u}_{\mathrm{cum},t}\Big) + C_{\mathrm{hedge},t}(\alpha_t) + C_{\mathrm{fill},t}(\alpha_t) + R_{\mathrm{safe}} \le B_{\mathrm{rem},t}, \tag{12}$$

where $s(\alpha)$ scales admitted shocks. Then for every disturbance path within the declared envelopes, the buffer after the step is at least R_{safe}; by induction, solvency holds at every step at which an admissible α exists. At $\alpha = 0$ the one-step admissible shock is

$$\bar{u}_{\mathrm{adm},t} = \frac{B_{\mathrm{rem},t} - R_{\mathrm{safe}} - |I_t|(1+\mu)\|z_t\|_2}{|I_t|(1+\mu)}.$$

The condition is sufficient over the whole disturbance ball. It is also necessary when the current state contributes no drift ($z_t = 0$), because the shock term is then attained by adversarially signed disturbances; with $z_t \neq 0$ the drift term $M_C\|z_t\|_2$ is an upper bound and the condition may be conservative. The admissible shock at $\alpha = 0$ assumes, as in the cohort economy (Lemma 3.3), that the price-row gain without admitted flow is $M_C(0) = 1$.

Two features matter. The guard accounts for the inventory the dealer will hold after hedging and after worst-case fills, not only current inventory, and it never claims safety on an infeasible state: if no α satisfies (12), it halts new fills and flags the state, because pre-existing inventory can still breach margin on a large adverse jump. In a constructed scenario ($E_0 = 10{,}000$, $I_0 = -2{,}000$ at $P^* = 100$, Tier-2 margin with $\mu = 0.005$, buffer 9,050 and reserve 950), the admissible shock at $z_0 = 0$ is $8{,}100/2{,}010 = 4.03$ per unit, and an

adversarial move of 4.05 violates the reserve.

Proposition 5.6 (When the guard beats an eigenvalue rule)**.** *Let an eigenvalue guard admit* $\alpha_\rho = \max\{\alpha : \rho(J(\alpha)) \le \bar{\rho}\}$*, and suppose the state is feasible for the summed-gain guard, which then admits at least* $\alpha = 0$*.*

(i) Over the Euclidean state ball $\{\tilde{z}_t : \|\tilde{z}_t\|_2 \le \|z_t\|_2\}$ *and one-step shock envelope* $|u_{t+1}| \le \bar{u}_{\text{step},t}$*, the eigenvalue guard violates the robust solvency bound* (12) *if and only if*

$$M_C(\alpha_\rho)\|z_t\|_2 + s(\alpha_\rho)|g_0(\alpha_\rho)|\bar{u}_{\text{step},t} \;>\; \frac{B_{\text{rem},t} - R_{\text{safe}} - C_{\text{hedge},t}(\alpha_\rho) - C_{\text{fill},t}(\alpha_\rho)}{|I^{\text{post}}_{t+1}(\alpha_\rho)|(1+\mu)}.$$

(ii) At a specific state z_t*, the eigenvalue guard admits an exact reachable one-step breach that the summed-gain guard prevents if and only if the exact reachable price excursion* $|c^\top_{\text{price}} J(\alpha_\rho) z_t| + s(\alpha_\rho)|g_0(\alpha_\rho)|\bar{u}_{\text{step},t}$ *exceeds the right-hand side above.*

The eigenvalue rule fails exactly when the transient gain at its admitted level exceeds the loss the balance sheet can absorb, which Proposition 3.2 shows can happen at any spectral threshold.

Proposition 5.7 (Strategic re-optimization reinforces the guard)**.** *Suppose cohorts re-optimize their steady-state Kalman gains when the venue throttles admission, and throttling scales the variance of execution-driven innovations as* $\sigma^2_\zeta(\alpha) = \alpha^2\sigma^2_{\zeta,0}$ *while measurement noise* $\sigma^2_\nu > 0$ *is unchanged. Then the optimal gain* $\eta^*(\alpha)$ *is strictly increasing in* α*, so throttling lowers every cohort's adaptation rate. If the transient gain is non-decreasing in the adaptation rates, then* $M_C(J(\alpha, \eta^*(\alpha))) \le M_C(J(\alpha, \eta^*(1)))$*: re-optimization does not undo the guard.*

The first statement is a property of the Riccati root; the second is conditional on monotonicity of the gain in η, which we check numerically. Across 3,000 stable draws of the three-state book with momentum (adaptation rates on a grid from 0.05 to 0.95), the peak response is non-decreasing in η in 98.8% of draws and in 99.9% of those with trend-following cohorts ($q > 0$); the summed absolute response is non-decreasing in 93% of draws with $q > 0$ but decreasing when cohorts are contrarian ($q < 0$). For trend-following cohorts, the case the filtering microfoundation produces, re-optimization therefore reinforces the guard in almost every case we drew; the signed sum S of Proposition 4.8 does not depend on η at all

(`recompute_oct20.py`). A manipulator who instead increases order size to offset throttling enlarges the shock envelope and drives α toward zero.

6 Who provides reversion? An equilibrium

Proposition 4.8 says that transient risk is governed by one ratio, $S = \tilde{\kappa}/(m - q\tilde{\kappa})$. Its ingredients are chosen by market participants: dealers decide how hard to lean against price displacements, which sets reversion m, and momentum traders decide how strongly to chase them, which sets q. This section asks whether those choices are efficient.

6.1 Setup

There are N dealers. Dealer i supplies reversion intensity $s_i \geq 0$ by absorbing order flow against the displacement, so $m = m_0 + \sum_i s_i$, where m_0 is reversion supplied by other means. Leaning earns a spread v per unit of intensity at quadratic inventory cost, so the private value is $vs_i - \frac{c}{2}s_i^2$. By Theorem 5.1 the worst-case transient loss of a position of size Q facing flow shocks bounded by U is $QU\,S$, so we take aggregate expected tail loss to be $\Lambda S(m, q)$ with $\Lambda > 0$. Dealers bear a share $\theta < N$ of it, so each bears θ/N; leveraged users and their counterparties bear the rest. Momentum traders choose aggregate intensity q with private value $bq - \frac{e}{2}q^2$ and bear none of the tail loss. Long-run impact $\tilde{\kappa}$ is taken as given.

6.2 Results

Proposition 6.1 (Equilibrium transient risk)**.** *Let* $g(m) \equiv -\partial S/\partial m = \tilde{\kappa}/(m - q\tilde{\kappa})^2$.

(i) Under-provision of reversion. *The symmetric Nash equilibrium solves* $m - m_0 = Nv/c + \theta\Lambda g(m)/c$*, the planner's optimum solves* $m - m_0 = Nv/c + N\Lambda g(m)/c$*, and* $m^{\mathrm{eq}} < m^{\mathrm{SP}}$*, so* $S^{\mathrm{eq}} > S^{\mathrm{SP}}$.

(ii) Fewer dealers, more risk. $\partial m^{\mathrm{eq}}/\partial N = (v/c)/(1 - \theta\Lambda g'(m)/c) > 0$*, so equilibrium reversion rises and* S^{eq} *falls with the number of dealers.*

(iii) Over-provision of momentum. *Momentum traders choose* $q = b/e$*; the planner chooses* $q^{\mathrm{SP}} < b/e$*, solving* $b - eq = \Lambda\tilde{\kappa}^2/(m - q\tilde{\kappa})^2$.

(iv) The corrective price is identified. *The first best for reversion is implemented by a rebate per unit of reversion*

$$\tau^* = \Big(1 - \frac{\theta}{N}\Big)\Lambda\, g(m) = \Big(1 - \frac{\theta}{N}\Big)\Lambda\, \frac{S^2}{\tilde{\kappa}},$$

which depends on the market only through S and $\tilde{\kappa}$.

Proof. (i) Dealer i's first-order condition is $v - cs_i + (\theta/N)\Lambda g(m) = 0$; the planner's is $v - cs + \Lambda g(m) = 0$. Summing over dealers gives the two fixed-point equations. The right-hand sides are decreasing in m because g is decreasing on $m > q\tilde{\kappa}$, and the planner's lies above the dealers' for every m when $\theta < N$, so its crossing with the line $m - m_0$ lies to the right. (ii) follows by implicit differentiation, since $g' < 0$. (iii) $\partial S/\partial q = \tilde{\kappa}^2/(m - q\tilde{\kappa})^2 > 0$ is the marginal external cost of momentum that traders ignore. (iv) The rebate closes the wedge between the two first-order conditions, and $g = S^2/\tilde{\kappa}$ by Proposition 4.8. □

At $v = 0.002$, $c = 0.5$, $\theta = 0.3$, $\Lambda = 10^{-4}$, $m_0 = 0.005$, $q = 0.02$ and $\tilde{\kappa} = 1$, equilibrium S falls from 27.9 with one dealer to 6.8 with forty, and the planner's S is 1.5–2.7 times lower throughout; the rebate computed from S alone coincides with the first-order wedge to machine precision. With $m = 0.03$, $b = 0.3$ and $e = 20$, momentum traders choose $q = 0.015$ against a planner's 0.0062.

Interpretation. Reversion is a public good: each dealer who leans against a displacement lowers the tail exposure of every leveraged position in the market but is paid only for his own fills. Momentum is the mirror image. Part (iv) closes the loop with the guard: the policy price of the externality is a function of the same summed response that the guard bounds, and Propositions 2.2 and 5.4 show that this quantity is identified from public data without identifying the traders. A venue that rebates reversion in proportion to $S^2/\tilde{\kappa}$, or a supervisor that sets pre-trade limits by S, prices the externality with an observable.

What the data say about the model's predictions. Part (ii) predicts that books with fewer dealers have higher S because they supply less reversion. Depth is the closest proxy we observe for the number of liquidity providers. Across the three liquidity terciles of the 30 Binance perpetuals on training days, the summed response per dollar of flow rises steeply as depth falls (median 2,797, 26,603 and 39,329 basis points per $1M in the thick, mid and thin

terciles; Spearman correlation with log depth -0.94), as (ii) predicts. The channel is not the one the model names, however: the unit-free gain multiple $S/|h_0| = 1/(m - q\tilde{\kappa})$, which isolates reversion net of momentum, is the same in all three terciles (medians 46.4, 47.3 and 46.9; Spearman -0.11). Thinner books carry more cumulative risk because their impact $\tilde{\kappa}$ is larger, not because they revert more slowly. These comparisons were made after the fact (Part 13, `R2`), and depth is an imperfect proxy for dealer count, so a direct test needs data on liquidity-provider identities. The model's second prediction, that the dynamic balance forecasts tail losses where reversion is close to $q\tilde{\kappa}$, is what Section 9 tests.

7 Evidence from Binance futures depth

7.1 Data and validation

We use Binance USD-M BTCUSDT perpetual futures from the public archive: one-minute klines with taker-buy volume, `bookDepth` snapshots of cumulative depth within ±1–5% of the mid at about 30-second intervals, and open interest from `metrics`. Depth is available from January 2023. No liquidation archive is available for this period, so we do not test liquidations.

Validation rules were fixed before analysis. A snapshot is kept if (V1) all of the $\pm 1, 2, 5\%$ bands are present, (V2) depth is monotone in distance on each side, (V3) the implied average price at ±1% is within 3% of the one-minute close, and (V4) it is not part of a run of more than ten identical snapshots; (V5) a day is kept if at least 80% of its minutes have a valid snapshot.

7.2 Structural estimates

On one-minute bars, p_t is the log price in basis points, u_t signed taker flow in million USDT, $d_t = (B_1 - A_1)/(B_1 + A_1)$ the depth imbalance within 1% of the mid, and $x_t = p_t - \mathrm{EMA}_{60}(p_t)$.

Day by day we estimate

$$\begin{aligned}\Delta p_t &= \kappa u_t + \varphi d_{t-1} - (1-a)x_{t-1} + e_t,\\ d_t &= (1-r)d_{t-1} + \varepsilon u_t + e_t,\\ u_t &= (1-\eta)u_{t-1} + \eta q\, x_{t-1} + e_t,\end{aligned}$$

and form $\hat{C}_2$, $\hat{C}_3$ and the implied peak ratio $\max_k \hat{h}_k/\hat{h}_0$. The price equation has Δp_t on the left, so the EMA filter does not create mechanical persistence. These are reduced-form regressions (flow is contemporaneous), so the estimates describe conditional dynamics rather than causal parameters.

Table 4 has three findings. First, the market sits on the stable side of the condition: $\hat{C}_3 < 0$ on 93.5% of days, and the median implied response peaks at impact. Second, the depletion coefficient $\hat{\varepsilon}$ is small and negative: taker buying is followed by slightly more ask-side than bid-side depth within 1%, so in these re-centred bands the book appears to replenish rather than deplete the side being hit; Section 9.4 shows that a fixed price grid reverses the sign. With $\hat{\varepsilon} < 0$, $\varphi\varepsilon$ is positive only if φ is negative, which it is not on high-volatility days. Third, on high-volatility days depth imbalance moves prices strongly ($\hat{\varphi} = 2.59$) and refill is twice as fast; the depth channel is active, but it works against amplification. The momentum term $\kappa^2\eta q$ is small (about 0.008 at the medians), far below $(1-a)\kappa \approx 0.022$.

7.3 Does depth carry an overshoot?

We estimate local projections (Jordà, 2005) of the cumulative price change on flow, $p_{t+k} - p_{t-1} = \beta_k u_t +$ controls, with five lags of returns, flow and depth: (A) total; (B) adding cumulative future flow, which removes order-splitting persistence; and (C) adding cumulative future depth imbalance, which removes the depth channel. Theorem 4.2 predicts an overshoot ($\max_{k\geq 1} \beta_k - \beta_0 > 0$) in (B) that shrinks in (C).

There is no overshoot (Figure 3, Table 5). The overshoot statistic is negative in every version. Holding depth fixed makes the response decay faster, consistent with depth that partly sustains the price move but never pushes it past impact.

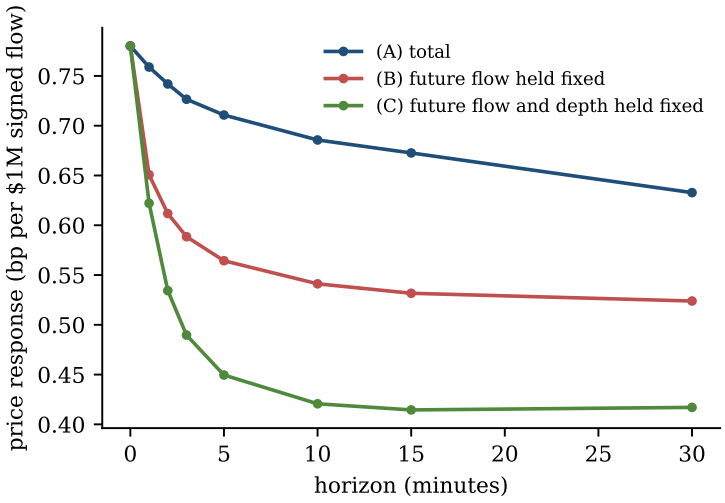


Figure 3: Price response to a $1M flow shock, local projections on 716 training days. (A) total; (B) future flow held fixed; (C) future flow and depth held fixed.

7.4 Pre-registered out-of-sample test

Before opening the test period we registered the specification (SHA-256 `d95212187ae4...`): hourly blocks (06:00–23:00 UTC); $\hat{C}_3$ estimated from the previous six hours and winsorized at training percentiles; outcome the next hour's maximum adverse excursion (MAE, bp); controls for realized volatility (1 and 6 hours), absolute return, flow imbalance, volume, depth, imbalance and open-interest change; OLS on 2023–2024 and evaluation on 2025–2026. Success required a one-sided Clark–West $p < 0.05$ *and* a day-block bootstrap 95% CI for ΔR^2 excluding zero. Secondary candidates and outcomes are Holm-adjusted. The test set was opened once. An audit that scrambles all data from t onward left every feature unchanged in 30 of 30 trials.

The primary test fails (Table 6): $\hat{C}_3$ adds 0.018 percentage points of out-of-sample R^2 with a confidence interval that includes zero. Some secondary effects are statistically significant after Holm correction, notably for depth collapse, but they are economically negligible: no candidate adds more than 0.2 points of R^2 or 0.01 of AUC.

A volatility flag does better. We set two flags on training data, each to flag 10% of hours: high $\hat{C}_3$ and high realized volatility. On the 909 tail hours of the test period (MAE

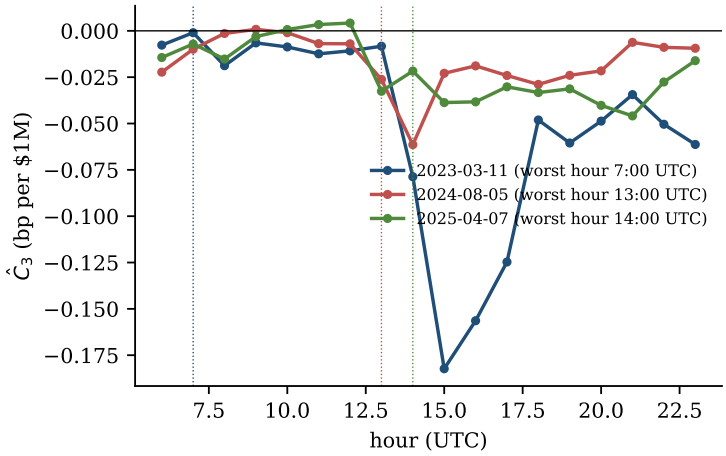


Figure 4: Hourly $\hat{C}_3$ on three event days (UTC). Values stay below zero, the amplification threshold.

above the training 90th percentile), the $\hat{C}_3$ flag has precision 8.6% and recall 19.5%; the volatility flag has precision 38.1% and recall 37.1%; the union has precision 16.3% and recall 49.4%. The unconditional tail rate is about 10%, so the $\hat{C}_3$ flag is no better than chance.

7.5 Crash events

For each event day we compute $\hat{C}_3$ hour by hour and its percentile in the training distribution during the three hours before the worst hour (Figure 4, Table 7). $\hat{C}_3$ was elevated before these events but stayed negative throughout, and three events are not a test.

8 Evidence from spot prices

8.1 A unit-free amplification statistic

The earlier version measured amplification by the norm of powers of a VAR(1) operator on (price deviation, flow), estimated over ten-minute windows of one-second Binance spot data. That norm depends on the units of the state: multiplying prices by 10 and dividing flow by 1,000 changed the raw norm from 8.73 to 86,731 on the same window. Proposition 2.2 explains why: state norms are not invariant to coordinates. We therefore standardize the

state, $\tilde{J} = D^{-1}\hat{J}D$ with $D = \mathrm{diag}(\mathrm{sd}\, x, \mathrm{sd}\, f)$, and use $M_{\mathrm{std}} = \max_k \|e_x^\top \tilde{J}^k\|$, which is invariant to units (verified). We also report the price-to-price amplification $A_{xx} = \max_k |(\hat{J}^k)_{11}|$ and a delayed-peak statistic hump $= \max_k |(\hat{J}^k)_{12}|/|\hat{J}_{12}|$.

8.2 Crash days and a no-feedback null

We study five crash days: May 19, 2021; LUNA, May 9, 2022; FTX, November 8, 2022; SVB, March 11, 2023; and August 5, 2024. Windows are selected *ex ante* (previous ten minutes in the top volatility decile), and for comparison by outcome (the ten largest absolute returns). A no-feedback null keeps each window's real flow, generates prices with no feedback, $\Delta p = \hat{\kappa} f + \hat{\sigma} e$, and pushes them through the identical pipeline.

Only LUNA exceeds the null, by less than 0.01% (Table 8). Outcome-selected windows give only marginally larger values (≤ 1.0005). Stress-window means exceed the ordinary 95th percentile in two events, by amounts of the same size. The pre-registered gate fails: unit-free transient amplification in spot data is indistinguishable from one and from the pipeline null. Varying the EMA span (60, 120, 300 seconds) and the bar length (1 or 5 seconds) does not change this; crash-day means range from 1.0003 to 1.18 across specifications and are never above the corresponding ordinary means.

8.3 Overshoot with flow held fixed

Local projections on one-second data give the price response to a \$1M flow shock (Table 9). On ordinary days, with future flow held fixed, there is a small, short-lived overshoot of +0.248 [0.133, 0.376] bp per \$1M, peaking at two seconds and decaying after. On crash days the flow-fixed overshoot is negative, −0.277 [−0.528, −0.042]. The large total overshoots (+0.91 ordinary, +2.40 crash) disappear when future flow is held fixed: they come from order splitting, not from the book.

8.4 Pre-registered predictive test

Before opening 2024, we registered (SHA-256 `b1fd7f8e...`) a test of whether pre-window amplification predicts the next ten minutes' MAE beyond volatility and flow controls, on non-overlapping 30-minute blocks (train 2021–2023, 3,307 blocks; test 2024, 1,104 blocks).

It does not: $\Delta R^2 = -0.023$ points $[-0.056, -0.002]$, Clark–West $p = 0.966$, AUC 0.822 in both models. A random feature with the same autocorrelation gives $\Delta R^2 = -0.020$ points, the same as the candidate. Leave-one-year-out fits within the training period are unstable (the 2023 fold gives a very large negative ΔR^2 because of outlying candidate values), so the null result does not depend on one year being unusual.

The test set was opened three times. The first evaluations stopped on a numerical error from degenerate windows; the fix drops blocks with non-finite features (five blocks) and is the only change, and the registered specification hash is unchanged. A volatility flag again dominates: precision 43% and recall 30% against 3% and 10% for the amplification flag; the difference in recall has a CI of $[-0.37, 0.13]$.

9 Evidence from thinner books

The deepest book in crypto is the least likely place to find amplification. This section moves to thinner books. It reports the pre-registered out-of-sample tests in thin Binance books, on Bybit and in thin US stocks, failures included, and labels every check decided after a test set was opened as post hoc. Figure 5 collects all out-of-sample evaluations of this section.

9.1 Design

We fixed 30 Binance USD-M perpetuals from the quote-volume ranking of one training day (June 10, 2024) before computing any T6 statistic, spread from BTCUSDT to symbols with under $5M daily volume. Data are sampled on the 10th and 24th of each month from January 2023 to September 2026 and pass the validation rules V1–V5 of Section 7: 2,565 of 2,700 symbol-days are valid; of 7,486,444 snapshots, V3 drops 212,745 and V4 drops 161,280, and V1 and V2 drop none. Liquidity terciles are fixed on median training-period depth within 1% of the mid. The thin tercile has $0.22M–$0.38M of depth within 1%, the thick tercile $0.97M–$220M. Flow is divided by each symbol's median absolute one-minute training flow, so that $\hat{\kappa}$ is in basis points per typical minute of flow. All other definitions follow Section 7. An audit that scrambles all data from t onward left every hourly feature unchanged in 30 of 30 trials.

9.2 The cross-section of the amplification condition

Table 10 confirms the BTCUSDT result across liquidity: mean $\hat{C}_3$ is negative in every tercile, and taker flow on average does not deplete the side it hits ($\hat{\varepsilon} < 0$ at the median everywhere). What changes with depth is the frequency of the amplifying regime, not its average. The share of symbol-days with $\hat{C}_3 > 0$ rises from 9.4% in the thick tercile to 16.1% in the thin tercile, with a Spearman correlation of -0.59 against log depth. Depth sensitivity $\hat{\varphi}$ is larger, and depletion is less negative and more often positive, in thinner books. We had predicted a negative correlation between log depth and mean $\hat{C}_3$; that prediction fails ($\rho = -0.001$).

The last row is an unexpected finding for the companion paper. Measured feedback $\hat{Q}$ is larger in *deeper* books ($\rho = +0.70$): flow responds to price deviations most where liquidity, and plausibly algorithmic participation, is greatest. Feedback and amplification are therefore different objects. Deep books have more feedback and refill fast enough to absorb it; thin books have less feedback and, more often, too little reversion.

9.3 Overshoot and depletion

With future flow held fixed there is no overshoot in any tercile (Table 11); the registered prediction of a positive (B) overshoot in thin books fails. The response decays less in thin than in thick books, and holding depth fixed again makes it decay faster, as in Table 5.

To check that replenishment is not an artefact of one-minute aggregation, we sum signed taker flow exactly between consecutive depth snapshots from trade-level data and estimate depletion ε_{30} at about 30 seconds, with an additional coefficient on sweeps (intervals in the top decile of absolute flow). Table 12 reports five symbols on six training days each.

In these re-centred bands, replenishment is stronger, not weaker, at 30 seconds: $\hat{\varepsilon}_{30}$ is negative with a CI below zero in three of five symbols and is never significantly positive. Sweeps offset most of it, so a sweep leaves the hit side about as deep as before ($\hat{\varepsilon}_{30}$ plus the sweep coefficient lies between -0.0012 and $+0.0007$). On this measure, the depletion channel that drives Theorem 4.2 would be absent even in thin books; the next subsection shows that the measure itself is biased toward replenishment.

9.4 Depletion on a fixed price grid

Binance's depth archive reports depth within bands around the *current* mid. After a buy moves the mid up, the ask band shifts into deeper resting orders and the bid band away from resting bids, which mechanically makes the book look ask-heavy after a buy, the sign we estimate. Bybit's order-book files contain every level, so the band can be held fixed at the previous mid. After seeing Table 12 we computed, for the six coins of Table 26 on four days each, the change in depth imbalance over 30-second intervals regressed on signed taker flow, once with the band fixed at the start-of-interval mid and once re-centred at the end-of-interval mid, Binance's construction.

The sign reverses (Table 13). With the band fixed, buying depletes the ask side in 19 of 20 coin-days; with the band re-centred, the same data reproduce the replenishment of Tables 4 and 12. The replenishment finding is therefore an artefact of band recentring, and depletion is present. Two conclusions survive. First, local projections show no peak overshoot once future flow is held fixed (Table 11), so depletion is not large enough to satisfy $\varphi\varepsilon > (1-a)\kappa$ in the data we observe; second, the depth term carries no out-of-sample forecasting content (Tables 15 and 16). Both statements rest on Binance's re-centred depth measure, and a fixed-grid measure of $\varphi\varepsilon$ in the same books would require order-level Binance data that are not public.

9.5 Pre-registered test in thin books

Before opening 2025–2026 for these symbols we registered the specification (SHA-256 `5910b6b4d90a...`). It is identical to Section 7 except for three changes, all fixed in advance: the primary sample is the thin tercile; the regressions include symbol fixed effects; and the bootstrap resamples dates, with all symbols on a date kept together. The test cell was executed twice in the same session. The specification hash, the code and the data were unchanged between the two runs, and the evaluation is deterministic, so both runs give the numbers below; we report the count because the protocol requires it.

The primary test passes (Table 14). In thin books, $\hat{C}_3$ estimated from the previous six hours adds 1.2 points of out-of-sample R^2 for the next hour's maximum adverse excursion, beyond realized volatility, flow, volume, depth, imbalance and open interest, and raises the AUC for tail hours by 0.012. The effect is confined to thin books: in the mid and thick

terciles it is zero, and in the pooled sample the Clark–West test rejects but the ΔR^2 interval contains zero. $\hat{C}_3$ also helps classify hours with a large open-interest drop, but it predicts depth collapse *worse* than the baseline, again pointing away from the depth channel.

Flags. On training data we set a $\hat{C}_3$ flag and a volatility flag; 575 of 6,521 test hours (8.8%) are tail hours. The $\hat{C}_3$ flag fires in 8.9% of hours with precision 16.5% and recall 16.7%, about twice the base rate. The volatility flag, which fires more often (16.0%), does better (precision 23.4%, recall 42.4%), and the union reaches recall 51.1% at precision 19.5%. $\hat{C}_3$ is a useful addition to volatility in thin books, not a replacement for it.

9.6 Tick size

Tick-size changes, a difference-in-differences design around them, and post-hoc tick-size robustness checks of Table 14 are reported in the Internet Appendix (Table 25): the result survives controls for tick size, the removal of days traded at a non-training tick and the removal of any single symbol, and weakens when the six thin symbols that ever changed tick are dropped.

9.7 Out-of-sample replication on new symbols

A result found in ten symbols could be specific to them. We therefore drew 24 new symbols, seeded and before any analysis, from the 72 USD-M perpetuals that were not in the original 30 and whose training-day volume was at most 1.5 times that of the most liquid thin symbol. We applied the same validation (2,090 of 2,160 symbol-days valid) and classified symbols as thin with a cut fixed in advance at the boundary between the original thin and mid terciles ($0.39M of depth within 1%). Ten new symbols are thin (ACH, ARPA, CHR, GTC, HOT, IOTX, OGN, SPELL, T and ZEN, all USDT-margined) and 14 are not. A new specification (SHA-256 `6aa8e2c03d79...`) fixed the primary test identical to Table 14 and, as secondary tests, the three terms of C_3 separately, a unit-free ratio $A = (\varphi\varepsilon + \kappa^2\eta q)/((1-a)\kappa) - 1$ that is positive exactly when $C_3 > 0$, and the not-thin group. The test set was opened once.

The result replicates (Table 15). On symbols never used before, $\hat{C}_3$ adds 0.85 points of out-of-sample R^2 with an interval excluding zero, and improves tail classification. It again has no value outside thin books. The decomposition shows where the forecasting content

lies, and every sign matches Theorem 4.2 and Proposition 4.4. A larger momentum term predicts larger adverse excursions; a larger reversion term, which pushes C_3 down, predicts smaller ones; and the depth term, whose depletion factor is close to zero (Table 12), predicts nothing. In training data the reversion term is by far the largest component (median 0.033, against -0.0009 for momentum and -0.0003 for depth), so most of the variation in $\hat{C}_3$ is variation in impact-weighted mean reversion.

The unit-free ratio fails. Because the amplifying terms are small, its median is -0.996 and its variation is mostly the noise of a ratio with an estimated denominator. Together with R5 in Table 25, this means that on these samples the forecasting content was tied to the level of $\hat{C}_3$. Section 9.9 tests two better-behaved unit-free statistics on fresh data.

9.8 Is $\hat{C}_3$ just price impact?

Both working terms contain the impact coefficient κ. In a post-hoc check on the ten new symbols, reported in the Internet Appendix (Table 23), $\hat{C}_3$ keeps about 80% of its gain in R^2 (0.68 of 0.85 points, interval above zero) with four impact controls ($\hat{\kappa}$, $\hat{\kappa}^2$, Kyle's λ and the Amihud ratio), both terms survive separately, and impact alone forecasts nothing; with $\hat{\kappa}$ controlled, the gain in tail classification is no longer significant. The later tests include these controls in the baseline.

9.9 A third test on fresh days

Parts 4 and 5 of the replication package use the 10th and 24th of each month. For a third test we registered a new specification (SHA-256 `65acda361861...`) before downloading the 3rd and 17th of each month of 2025–2026, days never used in any analysis. Training reuses the 2023–2024 days of the earlier tests. The sample is all 20 thin symbols of Sections 9.5 and 9.7, with the ten mid-tercile symbols as a negative control. Two changes make the test harder: the four price-impact controls of Table 23 are now part of the baseline, and because the specification has two primary hypotheses (this one and the backtest of Section 9.13) each is tested at $\alpha = 0.025$ with 97.5% intervals. The test set was opened once; 1,137 fresh symbol-days passed validation.

The result holds a third time, and more strongly (Table 16). On fresh days, with impact already in the baseline, $\hat{C}_3$ adds 1.6 points of R^2 for the next hour's adverse excursion in

thin books. Three findings sharpen the earlier picture. First, two unit-free statistics now work: the gain multiple, the summed response divided by impact, and the log half-life of reversion both add about half a point of R^2 with intervals above zero. The scale concern of Section 9.6 is therefore answered: the content is the dynamic balance, not the basis-point scale of the book, in the sense of the invariance hypotheses of Kyle and Obizhaeva (2016). Second, the reversion term carries most of the signal, as Proposition 4.8 suggests, since S depends on reversion through $1/(m - q\tilde{\kappa})$. Third, the depth-free statistic forecasts as well as the full one, which matters for the next subsection. In mid-depth books the Clark–West test rejects at the Holm-adjusted 5% level, but the R^2 gain is 0.06 points with an interval that includes zero. As with the impact controls, the gain in tail *classification* is not significant: the robust claim is about the size of excursions.

9.10 Beyond volatility forecasting

The adverse excursion of the next hour is close to a volatility outcome, so a natural objection is that $\hat{C}_3$ only proxies for future volatility in a way the simple volatility controls miss. We registered a fourth test (SHA-256 `71983f1a9d2c...`) on the 6th and 20th of each month of 2025–2026, days not used in any other test, whose baseline adds the standard tools of volatility forecasting to every control used so far: realized volatility over 15 minutes, one hour, six hours and since midnight in the spirit of HAR models (Corsi, 2009), an exponentially weighted (RiskMetrics) volatility forecast (J.P. Morgan/Reuters, 1996), the previous hour's adverse excursion, and hour-of-day fixed effects, together with the four price-impact controls. The test set was opened once; 1,161 fresh symbol-days passed validation.

The result does not weaken against a volatility-forecasting baseline; it strengthens (Table 17). $\hat{C}_3$ adds 2.8 points of out-of-sample R^2 on top of a baseline that already explains 18.6%, and for the first time it also improves the classification of tail hours with an interval excluding zero. The summed response S_{60}, the statistic that the guard bounds and that Proposition 4.8 identifies as cumulative transient risk, adds 2.1 points on its own. The volatility benchmarks, by contrast, add nothing out of sample beyond the controls of Table 16 (-0.76 points, interval $[-3.38, +2.10]$), so the forecasting content of the dynamic balance is not a volatility effect that richer volatility models would have captured. In mid-depth books the gain is significant in this test but small, about a ninth of the thin-book gain; across the

four tests with a deeper-book control it was significant once.

Comparison with VPIN. A related objection is that $\hat{C}_3$ proxies for order-flow toxicity as measured by VPIN (Easley et al., 2012). In a post-hoc check on the same 20 thin symbols and test hours of Table 17 (Part 13, `R8`), a 50-bucket six-hour VPIN statistic is negatively correlated with $\hat{C}_3$ in training data (Spearman $\rho = -0.35$) and adds +1.24 points of out-of-sample R^2 ($[+0.44, +2.30]$, Clark–West $t = 10.14$) to the full volatility-forecasting baseline, with no significant gain in tail classification ($\Delta\text{AUC} = -0.0007$). Adding $\hat{C}_3$ on top of the baseline plus VPIN yields an additional +2.03 points of out-of-sample R^2 ($[+1.01, +3.42]$, Clark–West $t = 12.14$) and +0.0039 of ΔAUC ($[+0.0014, +0.0063]$), retaining 73% of its unconditional +2.79-point gain. Flow toxicity and dynamic transient amplification capture distinct dimensions of thin-book risk.

9.11 A second venue: Bybit

To ask whether the result is specific to Binance we registered a test on Bybit USDT perpetuals of the same 20 coins (SHA-256 recorded in the package), using Bybit's public trade archive, which has no depth history. Fourteen of the coins have Bybit data across 2023–2026. Minute bars are built from trades (last price, signed taker notional), training uses the 10th and 24th of each month of 2023–2024, and testing uses the 3rd, 10th, 17th and 24th of each month of 2025–2026 (10,764 test hours). The statistic is $\hat{C}_3^{\text{flow}}$, which Table 16 shows to work on Binance.

The test fails without ambiguity. $\hat{C}_3^{\text{flow}}$ adds -0.0000 points of R^2 (95% CI $[-0.001, 0.001]$, Clark–West $p = 0.50$), and neither the momentum nor the reversion term helps (Holm $p = 1.00$). In training data the momentum coefficient even has the opposite sign (-130, against +110 on Binance). Because the depth-free statistic works on Binance on the same coins, the failure concerns the venue rather than the statistic. We considered two candidate reasons and checked the first after the fact: bid–ask bounce in Bybit's last-trade prices, and different participants. On the same 14 coins and training days, Bybit's one-minute returns are *less* negatively autocorrelated than Binance's (median -0.016 against -0.031), so bounce is not the difference; trading is: 11% of Bybit minutes have no trade, against 0.1% on Binance, and 29% of Bybit minutes have a zero return, against 14% (Part 13, `R3`). Sparse

trading leaves the flow and price series that the statistic is estimated from much noisier on Bybit.

Bybit with order-book depth. A second pre-registered Bybit test, with depth within 1% of the mid rebuilt from Bybit's order-book files and the full volatility baseline, kept six coins under its pre-registered inclusion rule. Its criterion is met ($\hat{C}_3$ adds 0.033 points of R^2, 95% interval $[0.019, 0.053]$, Clark–West $t = 5.73$), but the effect is about one eightieth of the Binance gain, tail classification does not improve, and only the momentum term helps. The Internet Appendix gives the design and the full results (Table 26). With trades alone the statistic has no content on Bybit, and with depth its content is detectable but too small to matter.

9.12 A regulated market: thin US stocks

The Bybit tests leave open whether the result belongs to thin markets or to one crypto venue. We therefore registered a test in US equities (SHA-256 `f5800a69c219...`) before downloading any test day. Data are consolidated (SIP) trades from the Alpaca market-data API, aggregated into minute bars of the regular session: last trade price, dollar volume and dollar flow signed by the tick rule. The data contain no depth, so the candidate is the depth-free statistic $\hat{C}_3^{\text{flow}}$, the version that worked on Binance (Table 16). The universe was drawn by a seeded rule from June 2024 daily bars, before any test data: 40 thin names (price \$5–\$100, median daily dollar volume \$2M–\$10M) and 10 mid-size controls (\$50M–\$200M). Forecasts are made 120, 180, 240 and 300 minutes after the open from the preceding 120 minutes, and the outcome is the maximum adverse excursion over the next 60 minutes; the baseline is the full volatility-forecasting and price-impact baseline of Table 17. Training uses 48 days in 2023–2024 (the 10th and 24th of each month or the next trading day) and testing 42 days from January 2025 to September 2026 (the 6th and 20th). A day is valid if at least 80% of its minutes have a trade, and a name enters if it has at least 30 valid training days and 25 valid test days, which keeps 23 of the 40 thin names and all 10 controls. The test cell was run once.

The test passes (Table 18). $\hat{C}_3^{\text{flow}}$ adds 0.50 points of out-of-sample R^2 to the full baseline (95% interval $[0.27, 0.78]$, Clark–West $t = 5.05$), and the summed response S_{60}, the guard's own statistic, adds 0.15 points. As on Binance, the content comes from impact-weighted

reversion: the reversion term alone adds 0.61 points, and the momentum term adds nothing. The effect is smaller than in thin Binance books, about a fifth of the 2.8 points of Table 17 on a baseline that already explains 34% of the variance, but about fifteen times the Bybit effect, and tail classification does not improve significantly. Unlike on Binance, the mid-size controls also show a gain (0.32 points), against our prediction of little or none: in US stocks the information is not confined to the thinnest names. The controls trade \$50M–\$200M a day, far below the deepest crypto books, and we included no large-capitalization stocks, so the test does not say where the effect ends. Three features of the sample limit the reading. The selection rule did not exclude funds, so the included thin names contain one closed-end fund (CRF) and the controls one bond ETF (BNDX); the validity rule removed the 17 thinnest-trading names; and the asset list contains only names still listed in 2026. Tick-rule signing also misclassifies some trades. Within these limits, the result is not a property of one crypto venue: the reduced-form content of the reversion statistic carries over, at a smaller size, to a regulated equity market.

9.13 The guard as a position limit

The guard of Section 5 bounds worst-case loss; a separate question is whether its ingredients help a dealer size positions in practice. We registered a backtest with Table 16. Each hour a rule sets the position $q_t = c_s/R_t$, where R_t is a risk measure from the past six hours and c_s sets the training mean of q to one for each symbol and rule; the hour's loss is q_t times the adverse excursion, and the score is tail loss per unit of average position, $\mathrm{TLR} = \mathrm{ES}_{95}(L)/\overline{q}$. The primary comparison is between a volatility limit (R = six-hour realized volatility) and the same limit scaled by the gain multiple to a power θ chosen on training data ($\theta = 0$ is the volatility limit); training data chose $\theta = 0.25$.

The primary backtest fails (Table 19). Scaling the volatility limit by the gain multiple lowers tail loss per unit of position in 91% of bootstrap draws, but the interval includes zero. Rules built only on the estimated response, without volatility, are clearly worse than the volatility limit and even than a static limit: a six-hour estimate of S is too noisy to size positions on its own. The guard theorems are statements about worst-case loss under a known or certified operator, and this evidence does not contradict them; but it means that, with public data and six-hour windows, an estimated summed response adds at most a little

to volatility as a sizing input. A practical guard should combine the two and certify the operator with more data than a six-hour window; Section 9.14 tests exactly that.

9.14 A certified guard

The guard theorems assume a certified operator, and a six-hour estimate is a poor certificate. We therefore registered a second backtest (SHA-256 `b9a56af48c93...`) in which the operator of each symbol is certified once a day by pooling the minutes of the previous three days, and applied to every hour of the following day. Evaluation days are the 28th of each month and certification windows the 25th to 27th; none of these days was used in any earlier test. Training is 2024 (12 evaluation days), testing January 2025 to September 2026 (20 evaluation days, 6,894 symbol-hours on the 20 thin symbols). The primary rule scales the six-hour volatility limit by the certified gain multiple, $R_t = \hat{\sigma}_{6h}\,\mathrm{GM}^{\theta}_{\mathrm{cert}}$, with θ chosen on training data; the score is the tail loss per unit of position of Section 9.13.

The primary test passes, narrowly (Table 20). With a certified operator, scaling the volatility limit by the gain multiple lowers tail loss per unit of position by 2.4%, with a 95% interval of $[-5.1\%, -0.04\%]$; training data again chose $\theta = 0.25$, as in Section 9.13. The gain is small and its interval almost touches zero, so we read it as a modest improvement, not a decisive one: amplification information helps as a correction to a volatility limit once the operator is estimated from days rather than hours. Every secondary comparison fails. A limit built on the certified summed response alone is far worse than volatility, about as bad as the six-hour version, so certification does not rescue the guard as a stand-alone sizing rule; and a combined ceiling that applies the stricter of the two limits does no better than volatility. The guard's theorems concern worst-case loss under a known operator; as a practical limit, its statistic earns its place as a modest adjustment to volatility, not as a replacement. The comparison also differs from the theorem in form: the backtests size positions as c/R_t, which levers a position up when the estimated response is small, whereas the guard of Theorem 5.1 is a one-sided cap, $\min(1, B/(IUS))$, that only scales admission down. After the fact we applied the certified guard in that form, throttling a position whenever $U_{\mathrm{cert}}S_{\mathrm{cert}}$ exceeds its training median (27% of test hours; Part 13, `R1`). As a cap on a static limit it lowers tail loss per unit of position by 13% (95% interval $[-27\%, -3\%]$); as a cap on a volatility limit it adds nothing ($+0.8\%$, $[-1.3\%, +2.7\%]$). Used as the theorem uses it, the guard helps where

no volatility information is used and is subsumed by volatility where it is.

9.15 October 10, 2025

The Internet Appendix describes what the statistics did on the largest liquidation day on record (Figure 7). The summed response was elevated all day and above its previous-week 90th percentile in 16 of 18 thin symbols at the crash hour, but volatility was equally extreme by then, and the statistic did not rank which coins fell furthest; the episode is descriptive and supports no inference.

9.16 Every pre-registered test

Table 21 lists every pre-registered test in this paper, with its outcome and the number of times its test set was opened. Of eleven tests, seven pass and four fail. The failures are in BTCUSDT futures, in BTCUSDT spot, on Bybit with trades alone, and in the six-hour position-sizing backtest. Four passes are in thin Binance books and one in thin US stocks; the certified-guard backtest passes narrowly; and the Bybit test with depth meets its criterion with a negligible effect. Figure 5 adds the post-hoc checks.

9.17 Threats to validity

Multiple testing. The program ran eleven pre-registered primary tests and the post-hoc checks of Figure 5. The six passing forecasting primaries have Clark–West statistics between 5.0 and 12.6 (one-sided $p < 0.001$ in each), so each survives a Bonferroni correction across all eleven primaries (threshold 0.0045); statistical significance is not the issue for the Bybit test, its size is. The certified-guard backtest does not survive the correction: its bootstrap one-sided p is about 0.02, so it is the weakest of the passes. The failures are reported with the same prominence.

Sequential design. The program was sequential. Each pre-registration fixed one test's specification before its test data were opened, but which test to run next was chosen after the earlier results were known: new symbols after the first pass, impact controls and then a volatility baseline in response to the obvious objections, depth on Bybit after the trade-only

test failed, a regulated market after the second venue gave a negligible effect, and a certified operator after the six-hour backtest failed. Pre-registration therefore protects each test from specification search on its own test data; it does not protect the program from choices that adapted across tests. That is why every test is reported, failures included (Figure 5), and why the correction above is taken over all eleven primaries.

Dependence among the passes. The four passing tests share the 2023–2024 training days, and the original ten thin symbols appear in three of them. They are not four independent replications. They vary three things one at a time: the symbols (Table 15, ten new symbols), the test days (Tables 16 and 17, days never used before) and the baseline (impact controls, then a full volatility-forecasting model). The US test (Table 18) shares none of their data: a different market, asset class, data source, universe and set of days.

Look-ahead and estimated regressors. Every feature uses minutes before the forecast hour; winsorization limits, flow scales, θ and tail thresholds come from training data; and a scrambling audit leaves every hourly feature unchanged when data from t onward are permuted. $\hat{C}_3$ and S_{60} are estimated from six hours, but the evaluation compares out-of-sample forecasts rather than testing a coefficient, so estimation error in the regressor can only attenuate the measured gain.

Simultaneity. The structural fits regress the price change on same-minute flow, so $\hat{\kappa}$ and the depletion slope mix impact with any within-minute response of flow to price. The guard's identification results (Propositions 2.2 and 5.4) concern the impulse response, not these regression coefficients, and the forecasting evidence uses $\hat{C}_3$ as an index whose value is judged out of sample, not as a causal parameter. Simultaneity therefore changes what $\hat{C}_3$ measures, not whether the out-of-sample gains are real. We tried to separate the two after the fact by instrumenting this minute's flow with last minute's flow (Part 13, `R4`), but at the one-minute scale normalized flow is almost serially uncorrelated (median first-stage correlation 0.005), so the instrument is weak and the instrumented statistic is noise (it adds 0.01 points of R^2, interval including zero). The causal content of $\hat{\kappa}$ therefore remains open; a convincing instrument would need order-level data.

Selection and survivorship. The 30 symbols were fixed from one 2024 training day before any T6 statistic was computed, and the new symbols were drawn by a seeded procedure. Symbols had to be listed through 2026, so coins delisted after a collapse are excluded; if anything this removes the episodes in which thin-book risk was largest.

Economic size. The largest gain, 2.8 points of R^2 on a baseline of 18.6%, is a 15% increase in the explained variance of the next hour's adverse excursion, and tail classification improves by 0.005 of AUC. In basis points, on the test hours of Table 17 and with quintile breakpoints from training data: within each quintile of the volatility forecast, hours in the top quintile of $\hat{C}_3$ have a next-hour adverse excursion 25.6 basis points larger than hours in the bottom quintile (date-block 95% interval $[18.6, 32.8]$), against a mean excursion of 112.5 basis points, and 61 basis points larger in the most volatile quintile; the share of hours beyond the training 95th percentile rises from 2.3% in the bottom $\hat{C}_3$ quintile to 5.1% in the top. Unconditional sorts on S_{60} show no pattern; its content appears only with the controls. As a sizing input the gain is worth about a 2.4% reduction in tail loss per unit of position once the operator is certified (Table 20). These are modest gains on top of a strong baseline, not a new predictor that replaces volatility.

In margin terms. To express the 25.6 basis points in the units a risk manager uses, we fit a lognormal distribution to the excursions in each $\hat{C}_3$ quintile from its mean and median; at one common threshold the fit reproduces the observed share of hours beyond the training 95th percentile in every quintile to within 0.6 percentage points. Shifting the bottom-quintile distribution up by 25.6 basis points, the effect at a given volatility forecast, roughly doubles the probability that a 20× position (liquidated after a 4% adverse move, with 1% maintenance margin) is liquidated within the hour, from 1.3% to 2.7% (2.3% to 3.2% across the interval for the effect). For a 10× position, liquidated after 9%, the probability almost triples, from 0.04% to 0.11%. On \$1M of notional at 20×, the extra expected loss of margin is about \$710 per hour held (\$490 to \$960; `recompute_dollar.py`). These figures come from a fitted distribution, not a test, and they treat the hour's excursion from its opening price as the liquidation trigger. They show that a difference that is small in basis points is large in the tail that leverage makes relevant, which is where a pre-trade control operates.

Venue. The size of the result depends on the venue. It is largest in thin Binance USD-M books; in thin US stocks it passes a pre-registered test at about a fifth of that size, and it also appears in mid-size US stocks (Table 18); on Bybit it failed with trade data alone and, with order-book depth, met its pre-registered criterion with an effect about one eightieth of Binance's (Table 26).

9.18 What the evidence says

Deep books sit on the stable side of the amplification condition. Flow depletes the side it hits on a fixed price grid, although re-centred depth bands make it look like replenishment, and we observe no peak overshoot. In thin Binance books the condition is still negative on average but binds more often, and the six-hour estimates of cumulative transient risk forecast the size of the next hour's adverse excursion out of sample. Four pre-registered tests pass, on the original symbols, on ten new symbols, on fresh days with price-impact controls, and on fresh days beyond a full volatility-forecasting baseline, where the gain is largest (+2.8 points of R^2) and extends to tail classification. The summed response S of Proposition 4.8, the guard's own statistic, carries the same information, and unit-free versions work. The content does not run through depth depletion, and in Binance books it is much weaker at mid depth and absent in BTCUSDT. In thin US stocks a fifth pre-registered forecasting test passes, at about a fifth of the Binance size and with a gain in mid-size stocks too; on Bybit the statistic is absent with trades alone and negligible with depth; and as a position-sizing input it improves a volatility limit only modestly once the operator is certified. Thin books are risky because their prices revert too slowly relative to the impact and momentum of the flow that moves them, which is what $S = \tilde{\kappa}/(m - q\tilde{\kappa})$ measures.

10 Policy implications

Pre-trade controls. Rule 15c3-5 requires pre-trade controls that prevent orders exceeding credit or capital thresholds (SEC, 2010). Static single-order caps and daily value-at-risk limits do not see transient dynamics. Theorem 5.1 identifies the quantity that a solvency-motivated control should bound: inventory times the shock envelope times the summed absolute impulse response. Peak-based or eigenvalue-based limits are either unsafe or blind

in exactly the regimes where they matter. Our evidence locates where such controls matter. In the deepest crypto book the amplifying regime is rare and the statistic carries no forecasting value; in thin books it is more frequent, and six-hour estimates of the summed response forecast the next hour's adverse excursion beyond volatility forecasts and price impact (Section 9). Our backtests also set a limit on what such a control can do with public data. A six-hour estimate of the summed response adds nothing to a volatility limit as a sizing input (Table 19); an operator certified from three days of data improves a volatility limit modestly, by 2.4% in tail loss per unit of position (Table 20); and limits built on the estimate alone do worse than volatility in both. The guard's role is therefore a ceiling on worst-case loss under a certified operator and a correction to conventional volatility limits, not a replacement for them, and it is most relevant for books with under about $0.4M of depth within 1% of the mid.

Pricing reversion. Proposition 6.1 shows that dealers under-supply reversion because its benefit, lower tail exposure for every leveraged position, is not paid to them. Maker rebates are a natural instrument: a rebate that rises with the book's summed response, $(1 - \theta/N)\Lambda S^2/\tilde{\kappa}$ per unit of reversion, prices the externality with an observable. Venues already tier rebates by volume; tiering them by contribution to reversion in thin books is a design question our model makes precise.

Market design. Frequent batch auctions remove latency races within a batch (Budish et al., 2015). Transient amplification, when it exists, operates over many seconds and spans many batches, so batching and a state-carrying guard address different failures and can be complements.

Supervision without identification. The companion paper shows that identifying algorithms from public data is expensive: any risk limit that depends on individual cohort memory rates η_i separately from the aggregate impulse response $(h_k)_{k\geq 0}$ is either unidentifiable from public prices under passive observation (where only the merged rate–weight measure $\mathcal{M}$ is identified) or inherits the ill-conditioned minimax sample complexity $(\Delta\eta)^{-(4K-2)}$ (and $(\log N)^{-1}$ for general completely monotone memory). By contrast, the guard needs only aggregate impulse coefficients and operator confidence sets, which are invariant to how flow

is split across cohorts (Propositions 2.2 and 5.4). When class-tagged disclosure is available, the companion paper's sample-efficiency gain of order $(\Delta\eta)^{-(4K-4)}$ shrinks the estimation confidence radius $\Delta G(\tau)$ in Theorem 5.3 by a factor of order $(\Delta\eta)^{-(2K-2)}$, reducing the robust admission haircut $\tau\Delta G(\tau)$ without active market probing (which would itself consume $\kappa\|u\|_{\ell_1}$ of the transient excursion budget). Supervisors can therefore ask venues and brokers to report summed-gain statistics and their confidence radii instead of attempting to infer black-box learners. Our results argue for reporting impact, mean reversion and refill directly, because those are the primitives that decide whether a book can amplify. They also show that feedback and amplification are different objects: measured feedback is largest in the deepest books, where amplification is rarest (Table 10).

11 Conclusion

The theory answers three questions. A book amplifies when depletion outweighs impact-weighted reversion; with momentum the one-period margin is sufficient but not exact, and momentum alone cannot make a stable book overshoot. Cumulative risk is long-run impact divided by reversion net of momentum, and the guard that bounds it is the largest safe pre-trade rule, computable and identified from public data. In equilibrium reversion is under-supplied, and the corrective price is a function of the guard's own statistic.

The evidence is mixed in an informative way. The deepest crypto book sits on the stable side of the condition; flow depletes the side it hits on a fixed price grid, but not enough to produce a peak overshoot in the data we observe. In thin Binance books, six-hour estimates of cumulative transient risk forecast the size of the next hour's adverse excursion out of sample in four pre-registered tests, in unit-free form, beyond price impact and beyond a full volatility-forecasting baseline. At a given volatility forecast, hours in the top quintile of the statistic have next-hour excursions about 26 basis points larger than hours in the bottom quintile. In thin US stocks, with consolidated trades, the statistic passes a fifth pre-registered test at about a fifth of the Binance size, and appears in mid-size stocks too. On Bybit it is absent with trades alone and negligible with order-book depth, the effect is much weaker in deeper Binance books, and as a sizing input it helps only once the operator is certified from days of data, and then modestly. Thin books are risky because their prices revert too slowly relative to the impact and momentum of the flow that moves them.

Three extensions would settle what remains open. Message-level data from a regulated market, such as LOBSTER or TAQ for US equities, would extend the US test to depth and to large stocks, show where the effect ends, and allow depletion to be measured without the recentering of band depth. Data on liquidity-provider identities would test the equilibrium's prediction that fewer dealers mean more transient risk. And longer certification windows, or certification with order-level data, would show whether the modest gain of a certified guard over volatility limits can be made large.

Data and code availability. All data are public: Binance archives (https://data.binance.vision), verified against published checksums, and Bybit's order-book history (https://www.bybit.com/derivatives/en/history-data) and trade archive (https://public.bybit.com); US consolidated trades are from the Alpaca market-data API (https://alpaca.markets, free account). Pre-registered specifications, their SHA-256 hashes and test-set unlock counters are in the replication package. Each number is produced by a notebook cell or script that writes a result file (Appendix I).

Use of AI tools. The authors used Claude (Anthropic), an AI assistant, to write and debug analysis code, run checks of derivations and numerical results, and draft and edit text. Every number in the paper is produced by code the authors executed and archived in the replication package. The authors reviewed all content and take full responsibility for it.

Disclosures. The authors declare no relevant financial interests.

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A Proofs

Proposition 2.1. With one cohort, $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$, so $\mathrm{tr}J = a+1-\eta+\eta\kappa q$ and $\det J = a(1-\eta)$. The Jury conditions for a real 2×2 matrix are $|\det J| < 1$ and $1 \pm \mathrm{tr}J + \det J > 0$. Substituting $a = 1-\kappa\varphi_f$: $1 - \mathrm{tr}J + \det J = \eta(1-a-\kappa q) = \eta\kappa(\varphi_f - q)$, positive iff $q < \varphi_f$; and $1 + \mathrm{tr}J + \det J = (4-2\eta) - \kappa[(2-\eta)\varphi_f - \eta q]$, positive iff the stated bound on κ holds (automatically when $(2-\eta)\varphi_f \le \eta q$).

Proposition 2.2. Equal merged measures give equal G, hence equal H, and the impulse coefficients are the Laurent coefficients of H (companion paper, Theorem on merged equivalence). Under $z \mapsto Tz$, $c^\top J^k b$ is unchanged because $CT^{-1}(TJT^{-1})^k Tb = CJ^k b$, while the row norm $\|CT^{-1}(TJT^{-1})^k\|$ depends on T. The displayed price path is the convolution of inputs with $\{g_k\}$, and its worst case over the ℓ_∞ ball is $\bar{u}\sum|g|$ by Hölder.

Theorem 3.1. (i) $\sup_t \|CJ^t z_0\| \le \sup_t \|CJ^t\| \|z_0\|$, with equality approached by aligning z_0 with the top right singular vector of CJ^{t^*} at the maximizing t^*. (ii) The Kreiss matrix theorem with Spijker's sharp constant gives $\mathcal{K}(J) \le \sup_t \|J^t\| \le en\mathcal{K}(J)$; substitute in (i). (iii) Take $J_\theta = \left(\begin{smallmatrix} \rho & \theta \\ 0 & \rho/2 \end{smallmatrix}\right)$; then $\rho(J_\theta) = \rho$ and $(1,0)J_\theta e_2 = \theta$, so $M_C \ge |\theta|$ and $r^*_{\text{exact}} \le B/|\theta| \to 0$.

Proposition 3.4. With all blocks entrywise nonnegative, every entry of J^t_{multi} is a polynomial in the entries of K_{21} with nonnegative coefficients, so it is non-decreasing in K_{21}; with $C \ge 0$, so is every entry of CJ^t and hence its norm. For the Kreiss constant, the Neumann series gives $|(zI-J)^{-1}| \le (|z|I-J)^{-1}$ entrywise for $|z| > 1$, so the supremum is attained on the real axis, where $(rI-J)^{-1} = \sum_k J^k/r^{k+1}$ is entrywise non-decreasing in K_{21}.

Proposition 4.3. The recursion (9) is solved by $h_k = Aa^k + Bb^k$ with $B(b-a) = \varphi\varepsilon$ and $A+B = \kappa$. Setting the derivative of $Aa^k + Bb^k$ in k to zero gives the unique critical point k^*; since the function rises at first ($h_1 > h_0$) and tends to zero, k^* is its maximum on $[0,\infty)$, and a unimodal function attains its maximum over the integers at $\lfloor k^* \rfloor$ or $\lceil k^* \rceil$. Summing (9) over $k \ge 1$ gives $S = \kappa + aS + \varphi\varepsilon/r$, so $S = (\kappa + \varphi\varepsilon/r)/(1-a)$, decreasing in r when $\varphi\varepsilon > 0$.

Theorem 5.5. After the step, inventory is at most $|I_{t+1}^{\text{post}}(\alpha_t)|$ in absolute value. The price move over the step and the cumulative window is bounded by $M_C\|z_t\|_2$ (drift from the current state) plus $s(\alpha)G_{\text{cum}}\bar{u}_{\text{cum}}$ (admitted shocks), by superposition and Hölder. The margin requirement scales the loss by $(1+\mu)$. Hedging cost and the post-fill margin increment are deducted directly. Inequality (12) therefore implies $B_{\text{rem},t+1} \geq R_{\text{safe}}$ for every admissible disturbance, and induction over steps at which an admissible α exists gives forward solvency. Setting $\alpha = 0$ ($M_C(0) = 1$, no fills, no hedge) gives $\bar{u}_{\text{adm}}$. Sufficiency over the Euclidean ball follows from the norm bounds. When $z_t = 0$ the remaining shock term is attained by $u = -\text{sign}(I)\bar{u}$, which gives necessity; when $z_t \neq 0$ the drift bound $M_C\|z_t\|_2$ need not be attained, so the condition is sufficient but not necessary.

Proposition 5.6. The eigenvalue rule admits α_ρ; a one-step breach occurs at α_ρ exactly when the left side of (12) with G_{cum} replaced by the one-step gain exceeds $B_{\text{rem},t}$, which rearranges to the stated inequality. The summed-gain guard admits only levels satisfying (12), so it never admits the breaching level, and by the feasibility assumption it admits at least $\alpha = 0$.

Proposition 5.7. The steady-state gain is $\eta = P/(P + \sigma_\nu^2)$ with P the positive root of $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$. Since P is strictly increasing in σ_ζ^2 and η is strictly increasing in P, η^* is strictly increasing in $\sigma_\zeta^2(\alpha) = \alpha^2\sigma_{\zeta,0}^2$, hence in α. The second statement is immediate under the stated monotonicity.

B Notation

Symbol	Meaning
$x_t,\ d_t,\ u_t$	price displacement, depth imbalance $(B - A)/(B + A)$, signed flow
$a,\ \kappa$	price persistence and direct impact
$\varphi,\ \varepsilon,\ r$	price sensitivity to depth, depletion per unit of flow, refill rate
$\eta,\ q$	momentum memory rate and aggressiveness
$C_2,\ C_3$	amplification statistics $\varphi\varepsilon - (1 - a)\kappa$ and $C_2 + \kappa^2\eta q$

Symbol	Meaning
$h_k,\ g_k$	impulse response of the book; price impulse coefficient of the controlled system
$J(\alpha)$	closed-loop operator at admission level α
$M_C,\ G_{\mathrm{cum}}$	peak and summed price-row gains
$\mathcal{K}(J)$	Kreiss constant
$\bar{u},\ \bar{u}_{\mathrm{step}},\ \bar{u}_{\mathrm{cum}}$	shock envelopes
$B_{\mathrm{rem}},\ R_{\mathrm{safe}},\ \mu$	remaining margin buffer, safety reserve, maintenance-margin rate
$\epsilon,\ S$	operator confidence radius and $\sum_k \|J^k\|$
$M_{\mathrm{std}},\ A_{xx}$	unit-free and price-to-price spot amplification statistics

Table 1: Map of results

Question	Answer	Where
Does $\rho(J) < 1$ bound transient loss?	No; safe radius B/M_C can vanish at fixed ρ	Thm 3.1, Prop 3.2
Can volatility monitors help?	No; fixed variance, unbounded gain	Prop 3.2
When does a book amplify?	Iff $\varphi\varepsilon > (1-a)\kappa$; with momentum $C_3 > 0$ suffices, no exact one-period rule	Thm 4.2, Prop 4.4, Rem 4.7
Can momentum alone amplify?	Not in a stable book	Thm 4.5
How large is cumulative risk?	$S = \tilde{\kappa}/(m - q\tilde{\kappa})$	Prop 4.8
Which guard is safe and largest?	Summed absolute gain; peak gain is unsafe	Thm 5.1
Robust to estimation error?	Yes, $M/(1 - \epsilon S)$	Thm 5.3
Needs identification of algorithms?	No; inputs are invariant to cohort splits	Props 2.2, 5.4
Do learners undo the guard?	No; gains fall when admission falls, and the peak response rises with gains in 99.9% of trend-following draws	Prop 5.7
Is reversion efficiently supplied?	No; under-provided, and the fix is priced by S	Prop 6.1
Is BTCUSDT on the amplifying side?	No: $\hat{C}_3 < 0$ on 93.5% of days	Table 4
Does depth carry an overshoot?	No; all overshoots negative	Figure 3
Does $\hat{C}_3$ forecast tail risk in BTCUSDT?	No (pre-registered)	Table 6
Is spot amplification above one?	No; unit-free gain ≈ 1.0000	Section 8
Does flow deplete thin books?	Yes on a fixed price grid; the apparent replenishment is a band-recentring artefact	Tables 12, 13
Does $\hat{C}_3$ forecast tail size in thin books?	Yes, in four pre-registered tests on Binance	Tables 14, 15, 16, 17
Is it just volatility?	No: +2.8 pts beyond HAR, EWMA and hour effects; S itself +2.1	Table 17
Is it tick size, impact or scale?	No; survives controls, unit-free versions work	Tables 25, 23, 16, 17
Does it hold on a second venue?	Bybit: absent with trades; with depth, significant but negligible	Section 9.11, Table 26
In a regulated market?	Thin US stocks: yes, +0.50 pts (pre-registered); mid-size stocks too	Table 18
Does the estimated response size positions better?	Six-hour: no; certified three-day: −2.4% tail loss, narrow pass; alone: worse	Tables 19, 20
How large is the effect?	top vs bottom quintile $\approx$ 26 bp at fixed volatility	Section 9.17

Table 2: Safe radius and Kreiss bounds at a fixed spectral radius

θ	$\rho(J_\theta)$	M_C (price row)	$\sup_t \|J_\theta^t\|$	$\mathcal{K}(J_\theta)$	r^*_{exact} $(B=1)$
0	0.8	1.00	1.00	1.00	1.000
1	0.8	1.36	1.37	1.06	0.735
5	0.8	6.03	6.04	3.43	0.166
20	0.8	24.01	24.01	13.42	0.042
100	0.8	120.00	120.00	66.99	0.008

$J_\theta = \left(\begin{smallmatrix} 0.8 & \theta \\ 0 & 0.4 \end{smallmatrix}\right)$; Kreiss constant by grid search over $|z| > 1$.

Table 3: Depth data validation

Days requested (January 2023 – October 2026)	1,386
Days passing validation	1,329
Snapshots examined	3,897,951
dropped by V1 (missing bands)	0
dropped by V2 (non-monotone)	0
dropped by V3 (implied price off by $> 3\%$)	98,080
dropped by V4 (stuck feed)	95,954
Failed periods	April 16 – May 19, 2025; October 10–15, 2025
Training days used (2023–2024)	718

All files verified against published SHA-256 checksums. The 2025 failures coincide with archived depth that is misaligned with prices; the October 10, 2025 crash day is therefore unusable.

Table 4: Day-by-day T6 estimates, Binance BTCUSDT futures, training days 2023–2024

Median	All days	Top-decile volatility days	Other days
$\hat{\kappa}$ (bp per \$1M)	0.729	0.890	0.712
$\hat{\varphi}$	−0.051	2.586	−0.138
$\hat{a}$	0.970	0.980	0.969
$\hat{r}$	0.059	0.122	0.056
$\hat{\varepsilon}$	−0.0018	−0.0017	−0.0018
$\hat{\eta}$	0.878	0.863	0.880
$\hat{q}$	0.018	0.012	0.019
$\hat{C}_3$	−0.0137	−0.0111	−0.0143
Mean $\hat{C}_3$	−0.0165	−0.0167	−0.0165
95% CI	[−0.0176, −0.0155]	[−0.0219, −0.0122]	[−0.0175, −0.0154]
Share of days $\hat{C}_3 > 0$	6.5%	5.6%	6.7%
Share of days $\hat{C}_2 > 0$	0.8%	1.4%	0.8%
Median peak ratio	1.00	1.00	1.00
Days	718	72	646

One-minute bars. CIs from a day bootstrap.

Table 5: Local-projection coefficients (bp per $1M) and overshoots

Minutes	0	1	2	3	5	10	15	30	Overshoot
(A) total	0.780	0.759	0.742	0.727	0.711	0.686	0.673	0.633	−0.021 [−0.027, −0.015]
(B) flow fixed	0.780	0.651	0.612	0.589	0.564	0.541	0.532	0.524	−0.129 [−0.134, −0.124]
(C) flow, depth fixed	0.780	0.622	0.534	0.490	0.450	0.421	0.414	0.417	−0.158 [−0.164, −0.152]

716 training days; day bootstrap.

Table 6: Out-of-sample test, 2025–2026 (train 12,896 hours, test 10,979 hours)

Outcome	Candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
MAE (primary)	$\hat{C}_3$	+0.018 [−0.100, +0.132]	+0.0001 [−0.0009, +0.0012]	—
MAE	peak ratio	+0.010 [+0.001, +0.019]	+0.0001 [−0.0000, +0.0002]	0.032
OI drop	$\hat{C}_3$	−0.045 [−0.095, −0.003]	−0.0049 [−0.0083, −0.0016]	0.787
OI drop	peak ratio	+0.102 [+0.061, +0.231]	−0.0014 [−0.0067, +0.0037]	< 0.001
Depth collapse	$\hat{C}_3$	+0.181 [−0.156, +0.484]	+0.0071 [+0.0041, +0.0105]	< 0.001
Depth collapse	peak ratio	+0.159 [+0.094, +0.234]	+0.0009 [+0.0004, +0.0014]	< 0.001

Primary: Clark–West one-sided $p = 0.023$, but the ΔR^2 CI contains zero, so the pre-registered test fails. Baseline R^2 for MAE is 0.293; baseline AUC 0.812.

Table 7: Event days

Event	Worst hour (UTC)	Worst MAE (bp)	$\hat{C}_3$ percentile, 3 hours before	OI change in worst hour
SVB, 2023-03-11	7	214.8	82	+2.5%
Yen carry unwind, 2024-08-05	13	724.0	93, 84, 83	−0.8%
Tariff shock, 2025-04-07	14	357.8	97, 98, 38	+3.1%
2025-10-10	no valid depth data			

SVB has one valid hour before its worst hour.

Table 8: Unit-free amplification on crash days and the no-feedback null

Event	Windows (ex ante)	Mean M_{std}	Null 99th pct	p (vs null)	Outcome-selected mean
May 2021	142	1.0000	1.00002	1.00	1.0000
LUNA 2022	57	1.00008	1.00006	0.039	1.00045
FTX 2022	49	1.0000	1.00029	1.00	1.0000
SVB 2023	6	1.00006	1.00025	0.059	1.00032
Aug 2024	108	1.0000	1.00007	1.00	1.0000

Ten-minute windows of one-second bars. The median hump statistic is 1.000 in every event and its null p-value is 1.00. In the ordinary 2021–2023 panel, 0.2% of windows have $A_{xx} > 1$ and the 95th percentile of M_{std} is 1.0000; a handful of near-singular windows produce very large values, so ordinary-window means are not informative.

Table 9: Spot local projections (bp per $1M)

Seconds	0	1	2	5	10	20	30	60	120	300
Ordinary, total	5.86	6.28	6.53	6.60	6.69	6.77	6.67	6.49	6.39	5.92
Ordinary, flow fixed	5.86	6.02	6.11	5.90	5.68	5.26	4.86	4.16	3.37	1.99
Crash, total	6.69	6.85	6.92	7.13	7.75	8.20	8.46	8.20	9.09	8.96
Crash, flow fixed	6.69	6.41	6.28	5.99	5.95	5.46	5.02	3.88	3.10	−0.08

Table 10: T6 estimates by liquidity tercile, 30 USD-M perpetuals, training days 2023–2024

	Thick	Mid	Thin
Mean $\hat{C}_3$	−0.0297	−0.0216	−0.0324
95% CI	[−0.0341, −0.0257]	[−0.0250, −0.0185]	[−0.0386, −0.0268]
Share of symbol-days with $\hat{C}_3 > 0$	9.4%	15.2%	16.1%
Median $\hat{\varphi}$	2.33	2.38	3.11
Median $\hat{\varepsilon}$	−0.0014	−0.0006	−0.0005
Share with $\hat{\varepsilon} > 0$	11.1%	24.4%	30.1%
Mean feedback loop gain $\hat{Q}$	0.0091	0.0033	0.0016
Spearman correlation with log depth across the 30 symbols [95% CI]			
mean $\hat{C}_3$		−0.001 [−0.37, 0.38]	
share $\hat{C}_3 > 0$		−0.59 [−0.81, −0.27]	
$\hat{Q}$		+0.70 [0.41, 0.87]	

1,424 training symbol-days, ten symbols per tercile. CIs from a symbol-day bootstrap (tercile rows) and a symbol bootstrap (correlations). $\hat{Q}$ is the loop gain of the companion paper estimated on one-minute bars; a placebo random walk with each day's volatility gives $\hat{Q} \approx 0$ for every symbol.

Table 11: Local-projection overshoot by tercile, $\max_{k\geq 1} \beta_k - \beta_0$ [95% CI]

	Thick	Mid	Thin
(A) total	−0.045 [−0.079, 0.047]	−0.034 [−0.054, 0.024]	+0.007 [−0.050, 0.083]
(B) future flow fixed	−0.181 [−0.211, −0.152]	−0.075 [−0.091, −0.058]	−0.073 [−0.094, −0.051]
(C) flow and depth fixed	−0.243 [−0.276, −0.212]	−0.121 [−0.140, −0.102]	−0.113 [−0.135, −0.092]
Symbol-days	459	460	445

Basis points per typical minute of flow, horizons 0–30 minutes; local projections are estimated per symbol-day and averaged, with a symbol-day bootstrap.

Table 12: Depletion at the snapshot scale (about 30 seconds)

Symbol	Tercile	$\hat{\varepsilon}_{30}$ [95% CI]	Sweep extra [95% CI]	$\hat{\varepsilon}$, 1 minute	Days $\hat{\varepsilon}_{30} > 0$
BTCUSDT	thick	−0.0045 [−0.0058, −0.0033]	+0.0034 [0.0023, 0.0046]	−0.0025	0 of 6
XMRUSDT	mid	−0.0042 [−0.0058, −0.0029]	+0.0037 [0.0025, 0.0051]	−0.0011	0 of 6
RLCUSDT	thin	−0.0034 [−0.0065, −0.0012]	+0.0029 [0.0011, 0.0058]	−0.0003	1 of 6
CELRUSDT	thin	−0.0015 [−0.0032, 0.0001]	+0.0017 [0.0004, 0.0031]	−0.0004	2 of 6
CVXUSDT	thin	+0.0010 [−0.0033, 0.0060]	−0.0003 [−0.0040, 0.0027]	+0.0004	2 of 6

Day bootstrap over six training days per symbol. Depth is measured in bands of ±1% around the current mid, so a price move also recenters the band; this biases $\hat{\varepsilon}_{30}$ in an unknown direction.

Table 13: Depletion at 30 seconds with a fixed and a re-centred band (Bybit order books, post hoc)

Band	$\hat{\varepsilon}_{30}$ [95% CI]	Share > 0	Coin-days
Fixed at previous mid	+0.0021 [+0.0013, +0.0032]	95%	20
Re-centred (Binance)	−0.0004 [−0.0007, −0.0001]	30%	20

ACH, ALICE, API3, LIT, WOO and ZEN, May 11–12 and September 11–12, 2025; flow normalized by each coin-day's median absolute 30-second flow; mean over coin-days with a bootstrap interval. Post hoc. Kind: real data (Part 13, `R5`).

Table 14: Pre-registered test, 30 symbols, 2025–2026

Sample	Outcome | candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	MAE | $\hat{C}_3$	+1.216 [+0.273, +7.987]	+0.012 [+0.003, +0.019]	—
Thin	MAE | peak ratio	+0.065 [−0.072, +0.108]	+0.000 [−0.000, +0.001]	0.207
Thin	depth collapse | $\hat{C}_3$	−2.111 [−3.127, −1.173]	−0.012 [−0.023, −0.003]	1.000
Thin	OI drop | $\hat{C}_3$	+0.184 [−0.242, +1.128]	+0.040 [+0.017, +0.061]	< 0.001
Mid	MAE | $\hat{C}_3$	+0.003 [−0.056, +0.875]	+0.002 [−0.002, +0.005]	0.551
Thick	MAE | $\hat{C}_3$	+0.039 [−0.244, +0.057]	−0.001 [−0.003, +0.000]	0.361
All 30	MAE | $\hat{C}_3$	+0.117 [−0.015, +1.736]	+0.003 [−0.000, +0.006]	< 0.001

Primary: train 8,340 and test 6,521 symbol-hours on ten symbols; Clark–West $t = 8.53$, one-sided $p < 0.001$; baseline out-of-sample $R^2 = 0.025$; AUC 0.768 → 0.780. The pre-registered test passes. Holm p is the Clark–West p-value adjusted across the six secondary tests. Date-block bootstrap with 2,000 draws; bootstrap CIs are right-skewed because a few volatile dates carry much of the variation.

Table 15: Pre-registered replication on ten new thin symbols, 2025–2026

Sample	Candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	$\hat{C}_3$	+0.845 [+0.303, +6.371]	+0.008 [+0.002, +0.013]	—
Thin	momentum term $\kappa^2\eta q$	+0.641 [+0.395, +3.242]	+0.003 [−0.001, +0.008]	< 0.001
Thin	reversion term $(1-a)\kappa$	+0.688 [+0.204, +5.606]	+0.008 [+0.003, +0.012]	< 0.001
Thin	depth term $\varphi\varepsilon$	+0.013 [−0.213, +1.155]	−0.001 [−0.005, +0.003]	0.308
Thin	unit-free ratio A	−0.045 [−0.096, −0.031]	−0.000 [−0.001, +0.000]	0.974
Not thin (14)	$\hat{C}_3$	−0.042 [−0.131, +0.780]	+0.002 [−0.003, +0.007]	0.308

Primary: train 8,520 and test 7,092 symbol-hours; Clark–West $t = 11.66$, one-sided $p < 0.001$; AUC 0.768 → 0.776; training coefficient +54.3 (Table 14: +88.3). Training coefficients: momentum +104.3, reversion −53.0. The pre-registered replication passes.

Table 16: Pre-registered test on fresh days, 2025–2026 (impact controls in the baseline)

Sample	Candidate	ΔR^2 (pts) [CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	$\hat{C}_3$	+1.583 [+0.588, +3.237]	+0.002 [−0.003, +0.006]	—
Thin	reversion term	+1.442 [+0.679, +2.621]	+0.003 [−0.001, +0.006]	< 0.001
Thin	momentum term	+0.422 [+0.000, +1.063]	−0.002 [−0.005, +0.001]	< 0.001
Thin	depth term	+0.323 [+0.092, +0.639]	+0.001 [−0.001, +0.003]	< 0.001
Thin	gain multiple $S_{60}/\lvert\hat{h}_0\rvert$	+0.488 [+0.252, +0.851]	+0.000 [−0.001, +0.001]	< 0.001
Thin	log reversion half-life	+0.619 [+0.351, +1.031]	+0.001 [+0.000, +0.003]	< 0.001
Thin	$\hat{C}_3$ without depth, $\hat{C}_3^{\text{flow}}$	+1.747 [+0.725, +3.393]	+0.001 [−0.004, +0.006]	< 0.001
Mid (control)	$\hat{C}_3$	+0.055 [−0.047, +0.239]	+0.000 [−0.001, +0.001]	0.038

Primary: train 16,860 and test 13,428 symbol-hours on 39 test dates; 97.5% date-block bootstrap interval; Clark–West $t = 9.46$, one-sided $p < 0.001$; baseline out-of-sample $R^2 = 0.185$; training coefficient +68.0. The pre-registered test passes. Secondary rows: 95% intervals, Holm-adjusted Clark–West p across the eleven secondary tests of the specification. Thin: 20 symbols; mid: 10. The gain multiple and the half-life are unit-free. S_{60} is the summed absolute impulse response over 60 minutes; $\hat{C}_3^{\text{flow}} = \hat{\kappa}^2\hat{\eta}\hat{q} - (1-\hat{a})\hat{\kappa}$ from the model without depth.

Table 17: Pre-registered test beyond a volatility-forecasting baseline, fresh days 2025–2026

Sample	Candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	$\hat{C}_3$	+2.787 [+1.425, +4.944]	+0.005 [+0.002, +0.008]	—
Thin	summed response S_{60}	+2.098 [+1.165, +3.597]	+0.004 [+0.001, +0.006]	< 0.001
Thin	reversion term	+2.451 [+1.257, +4.379]	+0.004 [+0.001, +0.007]	< 0.001
Thin	momentum term	+0.731 [+0.238, +1.429]	+0.001 [−0.001, +0.002]	< 0.001
Thin	gain multiple $S_{60}/\lvert\hat{h}_0\rvert$	+0.563 [+0.259, +1.003]	+0.001 [−0.001, +0.002]	< 0.001
Thin	log reversion half-life	+0.806 [+0.446, +1.351]	+0.001 [−0.000, +0.002]	< 0.001
Thin	$\hat{C}_3^{\text{flow}}$ (no depth)	+2.989 [+1.453, +5.424]	+0.005 [+0.002, +0.009]	< 0.001
Mid (control)	$\hat{C}_3$	+0.316 [+0.108, +0.635]	+0.002 [+0.001, +0.004]	< 0.001

Thin: 20 symbols, train 16,860 and test 13,722 symbol-hours on 40 test dates; mid: 10 symbols. Primary: Clark–West $t = 12.59$, one-sided $p < 0.001$; baseline out-of-sample $R^2 = 0.186$, full 0.214; AUC 0.826 → 0.831. The pre-registered test passes. Baseline: the controls of Table 16, the four impact controls, realized volatility over 15 minutes and since midnight, an EWMA volatility forecast ($\lambda = 0.94$), the previous hour's adverse excursion, hour-of-day fixed effects and symbol fixed effects. The gain multiple and the half-life are unit-free. Kind: real data.

Table 18: Pre-registered test in thin US stocks, 2025–2026

Candidate	ΔR^2 (pts) [95% CI]	Clark–West t	Holm p
Thin, $\hat{C}_3^{\text{flow}}$ (primary)	+0.499 [+0.272, +0.779]	5.05	—
Thin, summed response S_{60}	+0.153 [+0.046, +0.272]	3.36	0.001
Thin, reversion term	+0.613 [+0.322, +0.967]	5.15	< 0.001
Thin, momentum term	−0.011 [−0.288, +0.261]	0.63	0.263
Mid-size control, $\hat{C}_3^{\text{flow}}$	+0.316 [+0.074, +0.634]	3.12	0.002

23 thin names, train 3,744 and test 3,808 symbol-hours on 42 test dates; baseline out-of-sample $R^2 = 0.340$; AUC 0.846 → 0.847 (95% interval of the change $[-0.00002, +0.0026]$); training coefficient on $\hat{C}_3^{\text{flow}}$ +21.7. Mid-size controls: 10 names, 1,680 test symbol-hours. Specification SHA-256 `f5800a69c219...`; test opened once. Kind: real data (Part 14).

Table 19: Position-sizing backtest, thin books, fresh days 2025–2026

Rule	Risk measure R_t	TLR	Breach rate	$\log \text{TLR} - \log \text{TLR}_{\text{vol}}$ [CI]
Volatility limit	$\hat{\sigma}_{6h}$	348.6	1.9%	—
Amplification-adjusted (primary)	$\hat{\sigma}_{6h}\,(S_{60}/\lvert\hat{h}_0\rvert)^{0.25}$	342.7	2.1%	−0.017 [−0.045, +0.011]
Peak-response guard	$U_6 \max_k \lvert\hat{h}_k\rvert$	411.4	5.3%	+0.165 [+0.095, +0.243]
Summed-response guard	$U_6\, S_{60}$	522.2	13.2%	+0.404 [+0.285, +0.525]
Static	1	476.8	5.3%	+0.313 [+0.223, +0.397]

20 thin symbols; U_6 is the 95th percentile of absolute normalized flow over six hours. Primary row: 97.5% interval; others 95%. Breach: loss above the training 95th percentile of the static rule. Bybit: training chose $\theta = 0$, and the summed guard is worse than volatility by +0.484 [0.251, 0.662].

Table 20: Backtest with a certified operator, thin books, fresh days 2025–2026

Rule	R_t	TLR	Breach	$\Delta \log$ TLR vs vol [95% CI]
Volatility limit	$\hat{\sigma}_{6h}$	296.7	2.0%	—
Certified, amp.-adjusted (primary)	$\hat{\sigma}_{6h}\,\mathrm{GM}_{\mathrm{cert}}^{0.25}$	289.7	2.3%	−0.024 [−0.051, −0.0004]
Combined ceiling	stricter of the two	299.3	2.9%	+0.009 [−0.023, +0.039]
Certified summed guard	$U_{\mathrm{cert}} S_{\mathrm{cert}}$	416.5	13.8%	+0.339 [+0.241, +0.436]
Six-hour summed guard	$U_6 S_{60}$	424.8	14.9%	+0.359 [+0.237, +0.475]
Static	1	383.2	4.0%	+0.256 [+0.132, +0.387]

20 thin symbols; certified quantities from pooled within-day OLS on the minutes of the previous three days (median three valid days), fixed for the evaluation day. $\theta = 0.25$ chosen on training TLR over $\{0, 0.25, 0.5, 0.75, 1\}$. Combined ceiling: $R = \max(\hat{\sigma}_{6h}/\bar{\sigma},\, U_{\mathrm{cert}} S_{\mathrm{cert}}/\overline{US})$ with training medians. Breach: loss above the training 95th percentile of the static rule. Certified against six-hour summed guard: −0.020 [−0.165, +0.118]. Secondary comparisons Holm $p = 1.00$. Specification SHA-256 `b9a56af48c93...`; test opened once. Kind: real data (Part 12).

Table 21: All pre-registered tests

Test (specification hash)	Sample	Primary statistic		[CI]	Result	Opened
BTCUSDT futures, MAE | $\hat{C}_3$ (`d9521218`)	2025–26	ΔR^2	+0.018	[−0.100, +0.132]	fail	1
BTCUSDT spot, MAE | amplification (`b1fd7f8e`)	2024	ΔR^2	−0.023	[−0.056, −0.002]	fail	3
Thin books, 10 symbols (`5910b6b4`)	2025–26	ΔR^2	+1.216	[+0.273, +7.987]	pass	2
New thin symbols, 10 (`6aa8e2c0`)	2025–26	ΔR^2	+0.845	[+0.303, +6.371]	pass	1
Fresh days, 20 thin, impact controls (`65acda36`)	2025–26	ΔR^2	+1.583	[+0.588, +3.237]	pass	1
Position sizing, same specification	2025–26	$\Delta \log$ TLR	−0.017	[−0.045, +0.011]	fail	1
Bybit, 14 coins, depth-free	2025–26	ΔR^2	−0.000	[−0.001, +0.001]	fail	1
Beyond volatility forecasting, 20 thin (`71983f1a`)	2025–26	ΔR^2	+2.787	[+1.425, +4.944]	pass	1
Certified guard, position sizing (`b9a56af4`)	2025–26	$\Delta \log$ TLR	−0.024	[−0.051, −0.0004]	pass	1
Bybit with order-book depth, 6 coins (`ec2145fa`)	2025–26	ΔR^2	+0.033	[+0.019, +0.053]	pass	1
Thin US stocks, 23, depth-free (`f5800a69`)	2025–26	ΔR^2	+0.499	[+0.272, +0.779]	pass	1

ΔR^2 in percentage points. "Opened" counts executions of the test cell; the repeated openings used unchanged code and specifications (Sections 9.5 and 8). Intervals are 95%, except 97.5% for the two fresh-day primaries.

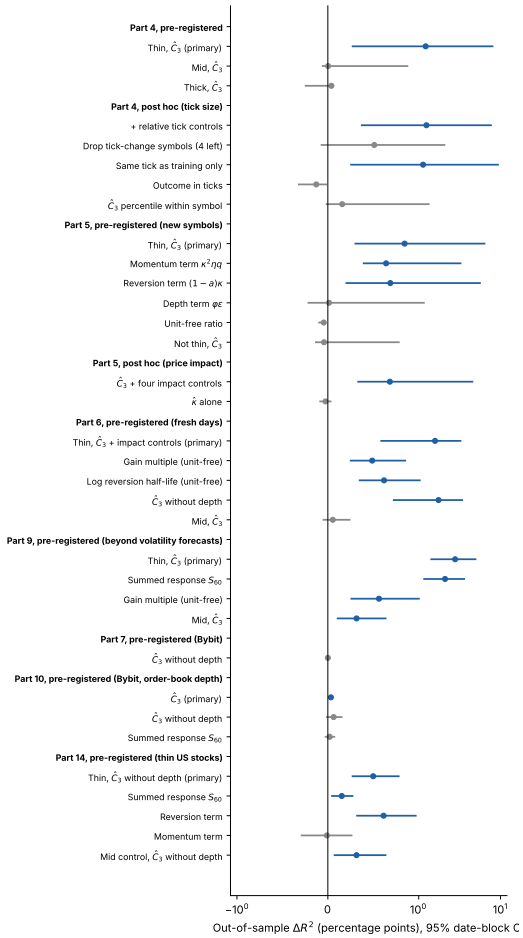


Figure 5: Every out-of-sample forecasting evaluation in this paper's thin-book program. Points are ΔR^2 in percentage points with date-block bootstrap intervals (symmetric-log axis). Blue: interval above zero. Pre-registered and post-hoc evaluations are grouped separately.

Table 23: Post-hoc price-impact controls, ten new thin symbols, 2025–2026

Candidate | added controls	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Clark–West t
$\hat{C}_3$ | none (Table 15)	+0.845 [+0.303, +6.371]	+0.008 [+0.002, +0.013]	11.66
$\hat{C}_3$ | $\hat{\kappa}$, $\hat{\kappa}^2$	+0.755 [+0.371, +4.718]	+0.006 [−0.000, +0.010]	6.63
$\hat{C}_3$ | Kyle's λ, Amihud	+1.089 [+0.498, +7.149]	+0.007 [+0.001, +0.013]	8.02
$\hat{C}_3$ | all four	+0.682 [+0.334, +4.525]	+0.004 [−0.001, +0.009]	5.69
Momentum term | all four	+0.487 [+0.311, +2.384]	+0.002 [−0.002, +0.006]	3.42
Reversion term | all four	+0.529 [+0.222, +3.762]	+0.005 [−0.000, +0.009]	8.45
$\hat{\kappa}$ alone | none	−0.026 [−0.084, +0.030]	+0.000 [−0.000, +0.000]	−1.77

Post hoc: decided after Table 15. Same evaluation code and bootstrap as Table 15; Holm-adjusted $p < 0.001$ for every row except $\hat{\kappa}$ alone ($p = 0.96$). Training correlations of $\hat{C}_3$ with the controls: $\hat{\kappa}$ −0.43, Kyle's λ −0.44, Amihud +0.10.

Internet Appendix for "From Human Markets to Machine Markets"

C Is $\hat{C}_3$ just price impact?

Both working terms contain the impact coefficient κ, so $\hat{C}_3$ might forecast excursions only because it measures illiquidity. After seeing Table 15 we added four impact controls, all from the same past six hours or hour and winsorized at training percentiles: $\hat{\kappa}$ from the same fit, $\hat{\kappa}^2$, a univariate Kyle's λ (Kyle, 1985) (slope of returns on signed flow) and a log Amihud ratio (Amihud, 2002) (absolute return over volume). This is a post-hoc check, evaluated once and counted separately.

$\hat{C}_3$ is not a proxy for impact (Table 23). With all four impact controls it keeps about 80% of its gain in R^2 (0.68 of 0.85 points), with an interval above zero, and both the momentum and the reversion terms survive separately. Impact on its own forecasts nothing. Controlling for Kyle's λ and Amihud alone *raises* the gain to 1.09 points: once plain illiquidity is held fixed, the dynamic balance that $\hat{C}_3$ measures stands out more clearly. One qualification: with $\hat{\kappa}$ controlled, the gain in tail *classification* is no longer significant. The robust claim is about the size of the next hour's adverse excursion, not about flagging tail hours beyond what impact already flags.

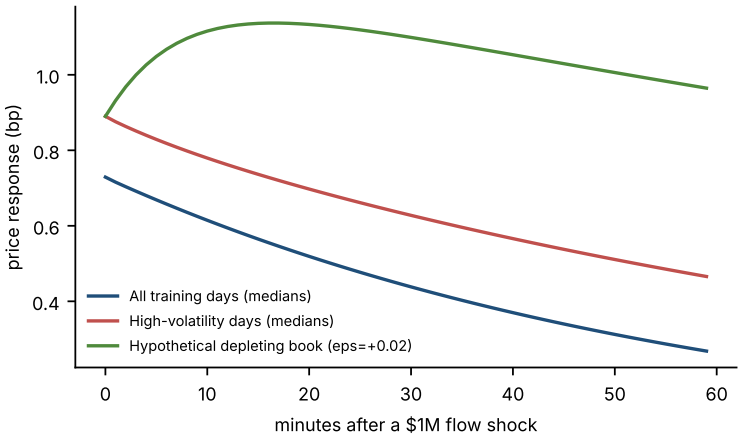


Figure 6: Price response to a $1M flow shock at the median depth estimates and in a hypothetical depleting book. Computed from the estimates of Table 4.

Table 24: Peak versus summed gain at calibrated parameters

Book	C_3	Peak (min)	$\sum_{k<15} \lvert g_k \rvert$	$\sum_{k<60} \lvert g_k \rvert$	Sum/peak	Cap, peak	Cap, sum
All days (medians)	−0.013	0.73 (0)	9.73	27.67	13.3	$274M	$21M
High-vol. days (medians)	−0.014	0.89 (0)	12.17	38.71	13.7	$225M	$16M
Depleting book (ε = 0.02)	+0.042	1.14 (17)	15.85	63.69	14.0	$176M	$13M

Gains in bp per $1M of flow; sum/peak at 15 minutes. Inventory cap under a peak guard and under the summed guard, for a loss budget of $100,000, an illustrative shock bound of $5M of flow per minute and a 15-minute horizon. From `calib_guard.py`, using the medians of Table 4.

D A calibrated illustration

How much does the choice between peak and summed guards matter in a real book? We compute the three-state impulse response at the median estimates of Section 7 (all training days and high-volatility days) and at a hypothetical book in which flow depletes the side it hits ($\varepsilon = +0.02$, other parameters at their high-volatility medians), Figure 6 and Table 24.

The result is striking even though BTCUSDT does not amplify. Because price impact is persistent ($\hat{a} \approx 0.97$ per minute), the summed gain over 15 minutes is about 13 times the peak. A guard calibrated to the peak would admit inventory about 13 times larger than is safe against a sustained adversarial flow of the declared size. Amplification makes this worse (the depleting book peaks 17 minutes after the shock), but persistence alone already makes

Table 25: Post-hoc tick-size robustness of the thin-book result

Check	Symbols	Test hours	ΔR^2 (pts) [95% CI]	Clark–West p
R0 baseline (Table 14)	10	6,521	+1.216 [+0.295, +8.017]	< 0.001
R1 + relative tick size and its log	10	6,485	+1.243 [+0.373, +7.627]	< 0.001
R2 drop symbols with any tick change	4	2,639	+0.511 [−0.069, +2.061]	< 0.001
R3 only days with the training tick	10	5,642	+1.131 [+0.256, +9.305]	< 0.001
R1 and R3 together	10	5,642	+1.222 [+0.357, +9.410]	< 0.001
R4 outcome in ticks	10	6,485	−0.127 [−0.319, −0.018]	0.991
R5 $\hat{C}_3$ percentile within symbol	10	6,521	+0.158 [−0.013, +1.322]	0.026
R6 leave one symbol out (10 fits)	9		+0.698 to +1.760, all CIs > 0	all < 0.001

Post hoc: decided after Table 14. 500 date-block bootstrap draws (200 in R6).

peak-based limits unsafe. This is the practical content of Theorem 5.1 for this market.

E Tick size: a caution and a robustness check

Tick changes. We located tick changes by inferring each day's price grid and dating changes by binary search over daily files. We found 12 tick reductions, all by a factor of ten, on 11 symbols (GMTUSDT twice) between August 2023 and August 2026; nine fall in the test period. A difference-in-differences design compares 10 days before and after each event with four control symbols of similar liquidity. After a tick cut, $\hat{C}_3$ rises with a CI above zero in 10 of 12 events, but realized volatility in basis points falls sharply at the same time (by up to 32 bp on ALICEUSDT), and the implied peak ratio does not change in any event (every CI includes zero). The Spearman correlation between the change in refill $\hat{r}$ and the change in the peak ratio is -0.04. We read this as a measurement result, not a test of the theory: at a coarse tick, prices move in discrete steps and the basis-point scale of $\hat{C}_3$ changes mechanically when the grid changes. The natural experiment cannot test the prediction that faster refill lowers the overshoot.

Does tick size drive Table 14? Eleven of the 30 symbols, six of them thin, changed tick, and five of those six did so inside the test period. After seeing Table 14 we therefore ran the post-hoc checks of Table 25, counted separately (one evaluation).

The result survives controls for tick size, the removal of every day traded at a tick different from training, and the removal of any single symbol (Table 25). It weakens when the six

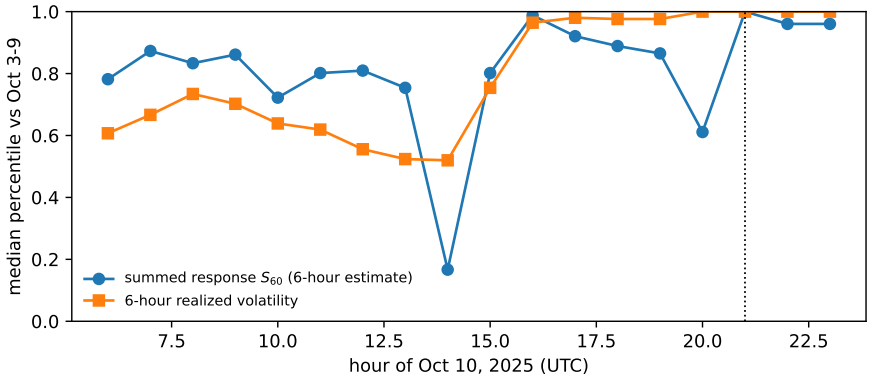


Figure 7: October 10, 2025: median percentile, across thin symbols, of the six-hour summed response S_{60} and of six-hour realized volatility, relative to each symbol's October 3–9 distribution. Dotted line: the crash hour (21:00 UTC). Descriptive; the event was chosen after the fact.

thin symbols that ever changed tick are dropped: four remain, and the point estimate halves with an interval that just includes zero. It disappears when the outcome is measured in ticks; at each tick cut this outcome jumps tenfold, a break the training fit cannot anticipate, so we regard R4 as a weak test, but we report it. A scale-free version of the candidate (its percentile within each symbol's training distribution) is weak. Our reading is that $\hat{C}_3$'s forecasting content is tied to its *level*, which mixes the amplification balance with the basis-point scale of each book. Sections 9.7 and 9.8 test whether that level is more than a measure of scale.

F October 10, 2025

The largest liquidation day on record motivated this paper, so we report what the statistics did on it, descriptively. The event was chosen after the fact and the 10th of each month was a test day in the first two thin-book tests, so this is not a test. For the 20 thin symbols we computed the hourly six-hour estimates on October 3–11, 2025, and expressed each hour of October 10 as a percentile of the same symbol's October 3–9 distribution. The crash hour, the hour on October 10 with the largest adverse excursion in BTCUSDT, began at 21:00 UTC; the median thin-symbol excursion in that hour was about 7,500 basis points.

Figure 7 shows two things. First, the summed response was elevated through the whole day: in the morning, before volatility rose, the median thin symbol's S_{60} sat at the 72nd to 87th percentile of its previous week, against the 52nd to 73rd for volatility, and at the start of the crash hour 16 of 18 thin symbols with valid estimates were above their previous-week 90th percentile. Second, volatility reached the top of its range from 16:00 onward as well, so on this day the summed response did not warn of anything volatility missed in the final hours; and across symbols it did not rank which coins fell furthest (Spearman -0.11, against $+0.36$ for volatility). We draw no inference from one day. The episode is consistent with the thin-book result, an elevated dynamic risk state that sizes excursions rather than timing them, and it illustrates why the paper's claims are about the size of the next hour's excursion, not about predicting crashes.

G Bybit with order-book depth

Bybit also distributes daily order-book files for its perpetuals (every update to the top 200 or 500 price levels). We registered a second Bybit test (SHA-256 `ec2145fac5f5...`) before downloading any test day, with the full Part 9 baseline, depth within 1% of the mid rebuilt from the order book at the end of each minute, and only the best 200 levels used so that the 2024 (500-level) and 2025–2026 (200-level) files are measured identically. Days were the 11th to 16th of a month, never used in any earlier test: two training months in 2024 and six test months from January 2025 to September 2026. A day is valid if at least 80% of its minutes have trades and a valid book; sparse trading in thin coins on Bybit leaves 581 of 882 coin-days valid, and the inclusion rule (at least 8 valid training days and 24 valid test days) keeps six coins (ACH, ALICE, API3, LIT, WOO, ZEN; 2,988 test hours on 36 dates).

The pre-registered criterion is met, but the effect is negligible (Table 26). $\hat{C}_3$ adds 0.033 points of R^2 with an interval above zero and a Clark–West t of 5.7: statistically clear, economically about one eightieth of the Binance gain, with no improvement in tail classification and a training coefficient one sixteenth as large. Of the components, only the momentum term helps; the reversion term, which carries the Binance result, does not. Two features of the sample may explain part of the gap, and we test neither: the six included Bybit books carry \$0.30M–\$0.43M within 1%, at or above the boundary between Binance's thin and mid terciles, where the Binance effect was also small; and the inclusion rule removed

Table 26: Pre-registered test on Bybit with order-book depth, 2025–2026

Candidate	ΔR^2 (pts) [95% CI]	Clark–West t	Holm p
$\hat{C}_3$ (primary)	+0.033 [+0.019, +0.053]	5.73	—
Momentum term	+0.083 [+0.017, +0.168]	2.94	0.010
$\hat{C}_3^{\text{flow}}$ (no depth)	+0.064 [−0.009, +0.152]	2.61	0.023
Summed response S_{60}	+0.021 [−0.025, +0.071]	1.31	0.383
Reversion term	−0.039 [−0.139, +0.058]	−0.61	1.000
Gain multiple	−0.043 [−0.137, +0.044]	−0.74	1.000
Log reversion half-life	−0.099 [−0.249, +0.041]	−1.47	1.000

Six coins, train 1,224 and test 2,988 symbol-hours; baseline out-of-sample $R^2 = 0.251$; AUC unchanged (0.808 → 0.808); training coefficient on $\hat{C}_3$ +4.4 (Binance, Table 17: +70.8). Median depth within 1% on training days \$0.30M–\$0.43M. The pre-registered criterion is met. Specification SHA-256 `ec2145fac5f5...`; test opened once. Kind: real data (Part 10).

the sparsest books. We read the two Bybit tests together: with trades alone the statistic has no content on Bybit, and with depth its content is detectable but too small to matter. Section 9.12 asks the same question in a regulated market.

H Additional material

H.1 Specification of the pre-registered tests

Futures (Table 6). Question: does $\hat{C}_3$, estimated from the previous six hours of futures depth, flow and price, predict the next hour's MAE beyond controls? Unit: hourly blocks 06–23 UTC, features from $[t-360, t)$ minutes, outcomes from $(t, t+60]$. Controls: rv_{60}, rv_{360}, $|r_{60}|$, flow imbalance, log volume, log depth, imbalance, open-interest change. Model: OLS; tail is the outcome above the training 90th percentile; score is the fitted value. Inference: Clark–West with Newey–West (6 lags); day-block bootstrap with 2,000 draws. Secondary: peak ratio as candidate; open-interest drop and depth collapse as outcomes; Holm. Train ≤ 2024-12-31; test ≥ 2025-01-01. Hash `d95212187ae42052591a5ca5acf73317 0f217f430a5a6ca812da6e48a1e4af85`; one unlock.

Spot (Section 8). Question: does pre-window A_{xx} predict the next ten minutes' MAE beyond volatility and flow controls? Unit: non-overlapping 30-minute blocks; features from the previous ten minutes. Controls: rv_{10m}, rv_{60m}, $|r_{10m}|$, flow imbalance, log volume, spectral radius, hour of day. Train 2021–2023; test 2024. Hash `b1fd7f8eef15df1ba295eb6d32b53be6c23ea2d23c5216de912555e62716e5ae`; three unlocks (see text).

I Reproducibility: source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part N" below and in table notes refers to notebook N (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–14: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard, post-hoc checks and US equities). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

Result	Source		Kind
Spectral blindness gains	`T5_spectral_blindness`		recomputed
Table 2; burst operator gain, q, certificate	`kreiss_table.py` → `R_P2_kreiss.json`	→	recomputed
Table 24, Figure 6	`calib_guard.py` → `R_P2_calibrated_guard.json`	→	recomputed
Figure 1	`make_figs2.py`		recomputed
Theorem 4.2, Propositions 4.3, 4.4 draws	`T6_amplification_theorem`		recomputed
Guard maximality; peak guard unsafe; monotonicity	`T7_guard_maximality`, `P2.11-T1`		recomputed
Robust gain, 30,000 draws	`T8_robust_gain`		recomputed
Operator ball coverage; certificate share	`P2.4-T1_coverage`		recomputed
Summed gain vs. weights	`T9_bridge`		recomputed
Depth validation (Table 3)	`D2_depth_validation`		real data

Result	Source	Kind
Table 4	`D3_T6_structural_train`	real data
Figure 3, Table 5	`D4_depth_mediation_train`	real data
No-look-ahead audit	`D5_no_lookahead`	audit
Table 6; hash	`D8_depth_test_G2`, `P2_depth.json`	real data
Flag comparison	`D8b_flag_comparison`	real data
Table 7, Figure 4	`D9_event_studies`	real data
Unit dependence 8.73 → 86,731	`P2.2-T1_unit_invariance`	real data
Table 8; sensitivity	`P2.2-T2`, `T3`, `T4`, `T5`	real data
Table 9	`T6-E_overshoot_real`	real data
Spot predictive test	`P2.5_predictive_G2`, `P2.5-T2`, `T3`	real data
Spot flag comparison	`P2.7-T2_vol_ablation`	real data
Proposition 4.8; Remark 4.7; Theorem 4.5 check	`equilibrium.py`, `t6m_search*.py`, `thmA_check.py`	recomputed
Proposition 6.1 numbers	`equilibrium.py` → `R_P2_equilibrium.json`	recomputed
Proposition 2.2 check; multi-asset example; Proposition 5.7 check	`recompute_oct20.py`	recomputed
Tables 10, 11, 12; tick events	Part 4: `X3`, `X4`, `X8`, `X7`	real data
Table 14; flags	Part 4: `X6_xsec_test_G2`, `X6b`; `P4_xsec.json` (5910b6b4...)	real data
Table 25	Part 4: `X6r_robustness_posthoc`	real data (post hoc)
Table 15	Part 5: `Y6_replication_G2`; `P5_replication.json` (6aa8e2c0...)	real data
Table 23	Part 5: `Y8_kappa_control_posthoc`	real data (post hoc)

Result	Source	Kind
Tables 16, 19; equilibrium check	Part 6: `Z6_*`, `Z8`; `P6_fresh_days.json` (65acda36...)	real data
Table 20, Figure 7, economic size (Section 9.17)	Part 12: `Q6_certified_guard`, `Q7_oct10_event`, `E1_economic_size`; `P12_certified_guard.json` (b9a56af4...)	real data
Liquidation probabilities and dollar losses (Section 9.17)	`recompute_dollar.py` (from `E1_economic_size`) → `R_P2_dollar_translation.json`	recomputed
Table 26	Part 10: `V3A_orderbook_download`, `V4_build`, `V6_bybit_ob_test`; `P10_bybit_ob.json` (ec2145fa...)	real data
Table 13; throttle, Bybit-bar, instrument and tercile checks	Part 13: `R5`, `R1`, `R3`, `R4`, `R2`	real data (post hoc)
Table 18	Part 14: `A1_us_universe`, `A3_us_build`, `A4_us_blocks`, `A5_us_equity_test`; `P14_us_equities.json` (f5800a69...)	real data
Proposition 4.3 peak bracket; Remark 4.7; Propositions 4.8, 6.1 checks	`recompute_audit.py` → `R_audit_oct2026.json`	recomputed
Section 9.11	Part 7: `W6_bybit_test`; `P7_bybit.json`	real data
Table 17	Part 9: `U6_volbench_test`; `P9_volbench.json` (71983f1a...)	real data
Figure 5	`make_fig_forest.py`	real data